Federated Learning over Hierarchical Wireless Networks: Training Latency Minimization via Submodel Partitioning

Wenzhi Fang, Dong-Jun Han, and Christopher G. Brinton

***Abstract*—Hierarchical federated learning (HFL) has demonstrated promising scalability advantages over the traditional "star-topology" architecture-based federated learning (FL). However, HFL still imposes significant computation, communication, and storage burdens on the edge, especially when training a large-scale model over resource-constrained devices. In this paper, we propose *hierarchical independent submodel training* (HIST), a new FL methodology that aims to address these issues in hierarchical settings. The key idea behind HIST is a hierarchical version of model partitioning, where we divide the global model into disjoint partitions (or submodels) per round so that each cell is responsible for training only one partition of the model, reducing client-side computational/storage costs and overall communication load. We characterize the convergence behavior of HIST under mild assumptions, showing the impacts of several key attributes (e.g., submodel sizes, number of cells, edge and global aggregation frequencies) on the convergence. Building upon the theoretical results, we propose a submodel partitioning strategy to minimize the training latency depending on the network resource availability (e.g., computation powers, data rate) and the learning performance constraint (i.e., convergence bound). We also propose an over-the-air computation version of HIST to further enhance efficiency during the model aggregation process over the hierarchical wireless network. Through simulations, we verify that HIST is able to save training time and communication costs by wide margins while achieving the same accuracy as the standard HFL baseline.**

***Index Terms*—Hierarchical federated learning, submodel training, wireless networks, over-the-air computation.**

I. Introduction

Massive amounts of training data generated by geographically distributed users have contributed to the huge success of modern machine learning (ML) applications. However, due to privacy and communication load issues, users may refuse to share their data with the service provider, which could potentially prohibit the implementation of centralized training. To avoid this issue, federated learning (FL) [2] was proposed as a promising distributed learning paradigm. Different from centralized training, in FL, distributed clients are responsible for training the model while the central server is only in charge of aggregating models of clients and disseminating the global model to clients. This enables clients to collaboratively learn a global model without any sharing of raw data. Owing to this benefit, FL has attracted significant attention in recent years [3]–[5].

In the traditional cloud-based FL [6], all clients in the system directly communicate with a central cloud server to exchange model information, resulting in communication scalability issues as the size of the network grows. To address this, hierarchical federated learning (HFL) has been proposed as an alternative [7]–[9], taking advantage of the fact that clusters of clients (e.g., cells) may be served by intermediate edge servers. The introduction of edge servers in HFL reduces communication and scheduling complexity, as the cloud server now only needs to communicate with the edge servers. However, as the size of the model increases, the HFL training process still suffers from scalability issues. These manifest in several dimensions: (i) computation/storage costs at individual clients, (ii) communication burden between clients and edge servers, and (iii) communication load between edge servers and the cloud server. These are fundamental bottlenecks for the practical deployment of HFL, especially when resource-constrained devices aim to collaboratively train a large-scale neural network model. This motivates us to investigate a communication-, computation-, and storage-efficient training algorithm for HFL.

In addition to the training methodology design for HFL, the deployment of HFL over wireless networks is also crucial. Due to the limited radio resources, the communication bottleneck of HFL at clients becomes more evident when it comes to wireless networks. For this reason, some researchers, e.g., [10], proposed to optimize resource allocation within each cell to facilitate learning performance. Additionally, the authors in [11] propose a joint client selection and resource allocation methodology to further enhance the efficiency of the radio resource. However, these works focus on orthogonal multiple access (OMA) transmission in each cell, in which the communication latency of each edge server linearly increases with the number of clients in its coverage. This issue further motivates us to study the implementation of over-the-air computation (AirComp) in HFL. By leveraging AirComp [12]–[14], we can expect that clients within the same cell can transmit their models simultaneously to the corresponding edge server, which will result in significant training time savings during the model aggregation process in each cell. The communication complexity at each edge server will become independent of the number of clients, thereby improving scalability over hierarchical wireless networks.

In this paper, we propose HIST, a new FL methodology

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that integrates *independent submodel training (IST)* in hierarchical wireless networks to relieve the bottleneck on resource-constrained clients and enhance the training efficiency. The core idea of HIST is to partition the global model into disjoint submodels in each training round and distribute them across different cells, so that devices in distinct cells are responsible for training different partitions of the full model. Such a submodel partitioning effectively reduces computation and storage loads at clients, and also alleviates communication burden on both the links between clients and edge servers and between edge servers and the cloud server. Additionally, given the limited spectrum resources and scalability issues in wireless environments, we propose an AirComp version of HIST to investigate the implementation of our methodology over hierarchical wireless networks.

A. Main Contributions

The main contributions of this paper are summarized as follows:

- We propose hierarchical independent submodel training (HIST) to facilitate FL over cloud-edge-client network topologies (Section II). HIST aims to reduce computation, communication, and storage costs incurred during HFL training via submodel partitioning across the edge.
- We analytically characterize the convergence characteristics of HIST (Section III) for non-convex loss functions, under milder assumptions than those adopted in the IST literature. Based on the convergence result, we analyze the performance-efficiency trade-off induced by submodel partitioning, and provide guidelines on setting the key system parameters of HIST.
- We analyze the impact of submodel size on the convergence bound and the training latency of HIST. Using these relationships, we propose a training latency minimization strategy (Section IV) which optimizes the submodel partitioning sizes without significantly compromising the learning performance (i.e., convergence bound), while considering the network resource availability (e.g., computation powers, data rate).
- To further enhance the training efficiency over wireless networks, we propose an AirComp version of HIST (Section V). With the assistance of AirComp, we show that each edge server is able to obtain an unbiased estimator of model averages of clients within its coverage. We characterize the variance of this estimate on the convergence behavior of the proposed AirComp-assisted HIST.

We conduct extensive experiments (Section VI) to validate the effectiveness of the proposed algorithm in hierarchical settings. Specifically, we train a fully connected neural network and a convolutional neural network (CNN) on Fashion-MNIST and CIFAR-10 datasets. Results show that HIST achieves significant resource savings for achieving the target accuracy compared with the standard hierarchical FedAvg [9]. Moreover, numerical experiments confirm that the optimized submodel partitioning strategy can further reduce the training latency without model performance degradation. We also show that, under a wide signal-to-noise ratio (SNR) region, the AirComp-assisted HIST algorithm attains almost the same testing accuracy as the OMA scheme while significantly reducing the training latency. To the best of our knowledge, this work is one of the earliest attempts to successfully integrate HFL, IST, and AirComp and to analyze its performance over wireless networks.

B. Related Works

Hierarchical federated learning: FL was first studied in a star-topology architecture where all clients are connected to a central cloud server [2]. The authors of [7], [8] extended FL to a hierarchical network that consists of a cloud server, edge servers, and clients, to reduce the communication complexity of the cloud server as well as the aggregation latency. Built upon this foundation, researchers have developed variants of the hierarchical FedAvg to further improve the training efficiency. Specifically, the authors of [15], [16], and [17] incorporated model quantization, sparsification, and compression into the standard hierarchical FedAvg to reduce the per-round transmission load. Furthermore, researchers in [10] focused on improving communication efficiency of hierarchical FedAvg by optimizing resource allocation during the model aggregation stage within each cell. Meanwhile, the authors of [11] formulated a joint client selection and resource allocation strategy to further enhance the efficiency when communication resources are limited. Note that the aforementioned works are orthogonal to ours where different groups train distinct partitions of the model via IST, and the techniques developed in these works can be extended to the proposed HIST idea.

Independent submodel training: The exploration of submodel training commenced with the pioneering work [18], where the authors introduced the concept of IST for fully connected neural networks and provided theoretical analysis under centralized settings. Subsequently, submodel training was extended to graph neural networks [19] and ResNets [20]. Due to its effectiveness in addressing communication, computation, and storage issues, the concept of IST was also considered in distributed scenarios [21], where the authors empirically show the effectiveness of submodel training in FL. Additionally, several studies also characterized the convergence behavior of distributed submodel training [22]–[24]. However, the aforementioned works either rely on restrictive assumptions [22], [23] or narrow the focus to quadratic models [24]. More importantly, existing works focus on cloud-based FL with a single server, and thus do not provide insights into the hierarchical case. To the best of our knowledge, HIST is the earliest work to integrate IST with HFL and provide theoretical analysis as well as experimental results.

AirComp-assisted FL: AirComp has been regarded as a promising solution for model aggregation in FL due to its efficiency. The authors of [25]–[27] proposed to utilize AirComp to accelerate the convergence of FedAvg in single-cell networks. Subsequently, the authors in [28] extended it to multi-cell FL and developed efficient interference management to minimize the inter-cell interference. In [29], researchers incorporated AirComp into HFL for edge model aggregation,

which successfully reduces the communication complexity of edge servers. More recently, the authors in [30] employed AirComp to support hierarchical personalized FL and characterized the impact of AirComp on the convergence by a non-diminishing term regarding aggregation distortion in the convergence bound. Despite the aforementioned works, there is still a lack of literature specifically addressing AirComp-assisted submodel training. Our contribution in this paper is to optimize submodel partitioning sizes with and without AirComp, and to provide guidelines on implementing IST over hierarchical wireless networks.

II. Hierarchical Independent Submodel Training Algorthm

In this section, we first outline the problem formulation of HFL, followed by the description of the proposed HIST algorithm tailored to HFL. Subsequently, we delve into the communication complexity analysis and highlight the efficiency of HIST.

A. System Model and Formulation

We consider a HFL system that consists of a single cloud server, N edge servers indexed by $\{1, 2, \ldots, N\}$, and $\sum_{j=1}^{N} n_j$ clients, where n_j is the number of clients located in the j-th cell. Edge server j is responsible for coordinating the training process of n_j clients in cell j. The global server is in charge of model aggregation over N geographically distributed edge servers. Given the loss function $l(\boldsymbol{x}, \xi_i)$ which measures the loss on sample ξ_i and model $\boldsymbol{x} \in \mathbb{R}^d$, the training objective of HFL can be formulated as follows:

$$\begin{aligned} \min_{\boldsymbol{x}} f(\boldsymbol{x}) &:= \frac{1}{N} \sum_{j=1}^{N} f_j(\boldsymbol{x}) && \text{Global loss} \\ f_j(\boldsymbol{x}) &:= \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} F_i(\boldsymbol{x}) && \text{Cell loss} \\ F_i(\boldsymbol{x}) &:= \mathbb{E}_{\xi_i \sim \mathcal{D}_i} \left[l(\boldsymbol{x}, \xi_i) \right] && \text{Client loss} \end{aligned} \tag{1}$$

where $f : \mathbb{R}^d \to \mathbb{R}$, $f_j : \mathbb{R}^d \to \mathbb{R}$, and $F_i : \mathbb{R}^d \to \mathbb{R}$ represent the global loss, the loss of j-th cell, and the loss of client i, respectively. $\mathcal{C}_j$ denotes the client set in cell j, $\mathcal{D}_i$ denotes the local data distribution of client i, and d is the number of parameters of the model. In this work, we mainly consider the non-i.i.d. scenario where data distribution is heterogeneous across different clients, i.e., $\mathcal{D}_j \neq \mathcal{D}_{j'}, \forall j' \neq j$.

In conventional HFL, all clients in the system are required to train the full model. To support such training, each client needs to be equipped with enough computation, storage, and communication resources. However, as the size of the models continues to grow, it is unaffordable for resource-constrained clients to handle training of large-scale models. This motivates us to develop a more efficient training framework for HFL, as we will discuss in the next subsection.

B. Algorithm Description

Inspired by IST [18], we develop a hierarchical federated submodel training algorithm termed HIST, by incorporating hierarchical FedAvg and submodel partitioning. The overview of HIST is presented in Fig. 1 and Algorithm 1. The cloud server periodically aggregates edge models from edge servers, while each edge server periodically aggregates local models from the clients within the corresponding cell. The key difference with the conventional HFL is that, clients do not need to store, update, and exchange the full model in HIST.

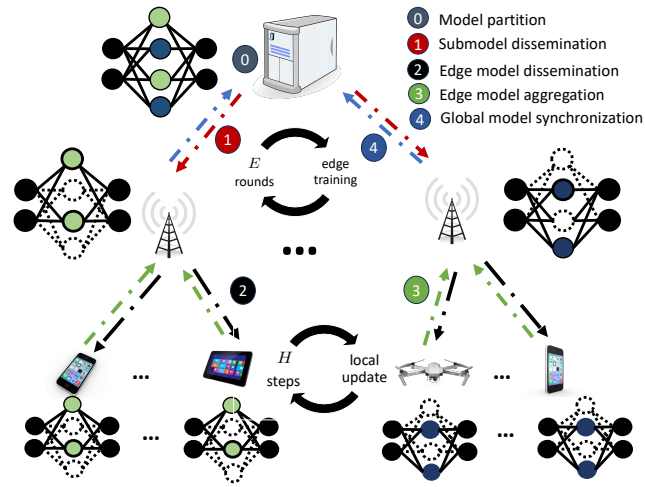


Fig. 1: Overview of proposed hierarchical independent submodel training (HIST). Each cell is responsible for training only a specific partition of the full model, where the submodel partitioning changes over global rounds.

Specifically, at the beginning of t-th global round, the cloud server initiates the training process by partitioning the current global model $\bar{\boldsymbol{x}}^t$ into N disjoint submodels as follows:

$$\{\boldsymbol{p}_j^t \odot \bar{\boldsymbol{x}}^t \mid j = 1, 2, \ldots, N\}, \tag{2}$$

where $\odot$ denotes the Hadamard Product operation and $\boldsymbol{p}_j^t$ is a mask that has either 0 or 1 in its element and satisfies

$$\boldsymbol{p}_j^t \odot \boldsymbol{p}_{j'}^t = \boldsymbol{0}, \forall j' \neq j, \text{ and } \sum_{j=1}^{N} \boldsymbol{p}_j^t = \boldsymbol{1}. \tag{3}$$

These submodels are then distributed to edge servers. Edge server j receives $\boldsymbol{p}_j^t \odot \bar{\boldsymbol{x}}^t$ from the cloud server and initializes its model as

$$\bar{\boldsymbol{x}}_j^{t,0} = \boldsymbol{p}_j^t \odot \bar{\boldsymbol{x}}^t, \forall j \in \{1, 2, \ldots, N\}, \tag{4}$$

where 0 in the superscript refers to the initial edge model at a specific global round t. Subsequently, edge server j disseminates $\bar{\boldsymbol{x}}_j^{t,0}$ to the clients within its coverage for local model initialization, i.e., $\boldsymbol{x}_{i,0}^{t,0} = \bar{\boldsymbol{x}}_j^{t,0}, \forall i \in \mathcal{C}_j$, where 0 in the subscript refers to the initial local model at a specific edge round. Once the clients receive the most recent model from the edge server, they start local training on their own data.

In our framework, each global round consists of E steps of edge aggregation at each edge server and one global aggregation at the cloud server. We denote (t, e)-th round as the e-th edge round at global round t. Moreover, each edge round includes H steps of local updates at each client and one edge aggregation at the edge server.

The essential steps conducted by clients, edge servers, and the cloud server in our algorithm are outlined as follows.

Clients: Each client i first computes stochastic gradients with respect to its corresponding submodel, and then updates the local model for H steps via the following iteration:

$$\boldsymbol{x}_{i,h+1}^{t,e} = \boldsymbol{x}_{i,h}^{t,e} - \gamma \boldsymbol{p}_j^t \odot \nabla l(\boldsymbol{x}_{i,h}^{t,e}, \xi_{i,h}^{t,e}), e = 0, 1, \ldots, H-1, \quad (5)$$

where γ represents the learning rate and t, e, and h denote the number of global rounds, edge rounds, and local iterations, respectively. Note that $\boldsymbol{p}_j^t$ keeps invariant during one global round. Subsequently, each client uploads the updated submodel to the corresponding edge server.

Edge Servers: After every H steps of local submodel updates at the clients, each edge server j aggregates the local models within its coverage as

$$\bar{\boldsymbol{x}}_j^{t,e+1} \leftarrow \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \boldsymbol{x}_{i,H}^{t,e}, \forall j \in \{1, 2, \ldots, N\}. \quad (6)$$

Subsequently, edge servers determine whether to upload the aggregated model to the cloud server or disseminate it to the clients. The criterion is whether each edge server has finished E rounds of edge training. If not, they disseminate edge models obtained in (6) to the corresponding clients; otherwise, each edge server uploads the aggregated model to the cloud server.

Cloud Server: Once the cloud server receives the latest models from all edge servers, it synchronizes these edge models and updates the global model as follows:

$$\bar{\boldsymbol{x}}^{t+1} = \sum_{j=1}^{N} \bar{\boldsymbol{x}}_j^{t,E}. \quad (7)$$

Subsequently, at the beginning of the next global round, i.e., $t+1$, the cloud server repartitions the global model $\bar{\boldsymbol{x}}^{t+1}$ based on newly generated masks $\boldsymbol{p}_j^{t+1}$ as

$$\bar{\boldsymbol{x}}_j^{t+1,0} = \boldsymbol{p}_j^{t+1} \odot \bar{\boldsymbol{x}}^{t+1}, \forall j \in \{1, 2, \ldots, N\}. \quad (8)$$

Finally, $\bar{\boldsymbol{x}}_j^{t+1,0}$ will be sent to edge server j for initiating the next round of training.

With the proposed algorithm, clients and edge servers are not required to store or manipulate the full model parameters. This enables HIST to alleviate the communication, computation, and storage burdens for clients and edge servers compared to the conventional HFL.

Algorithm 1: Hierarchical Independent Submodel Training Algorthm

Input: Initial masks $\{\boldsymbol{p}_1^0, \boldsymbol{p}_2^0, \ldots, \boldsymbol{p}_N^0\}$, initial models $\bar{\boldsymbol{x}}^0$, and $\boldsymbol{x}_{i,0}^{0,0} = \bar{\boldsymbol{x}}_j^{0,0} = \boldsymbol{p}_j^0 \odot \bar{\boldsymbol{x}}^0, \forall i \in \mathcal{C}_j, \forall j$, learning rate γ
for $t = 0, 1, \ldots, T-1$ **do**
  for $e = 0, 1, \ldots, E-1$ **do**
    for $h = 0, 1, \ldots, H-1$ **do**
      Clients update local model by (5) in parallel
    end for
    Edge servers update edge models $\bar{\boldsymbol{x}}_j^{t,e+1}$ by (6) in parallel
    Edge servers send the updated edge models to clients: $\boldsymbol{x}_{i,0}^{t,e+1} \leftarrow \bar{\boldsymbol{x}}_j^{t,e+1}, \ \forall i \in \mathcal{C}_j$
  end for
  Cloud server updates the global model $\bar{\boldsymbol{x}}^{t+1}$ by (7)
  Generate masks $\{\boldsymbol{p}_1^{t+1}, \boldsymbol{p}_2^{t+1}, \ldots, \boldsymbol{p}_N^{t+1}\}$ under rule (3)
  Partition the global model by (8) and send the obtained submodels $\bar{\boldsymbol{x}}_j^{t+1}$ to clients within cell j, $\boldsymbol{x}_{i,0}^{t+1,0} = \bar{\boldsymbol{x}}_j^{t+1}, \ \forall i \in \mathcal{C}_j, \forall j$
end for

C. Communication Complexity Analysis

Let L_0 denote the full model size. Assume that the mask size, defined as the number of non-zero entries of $\boldsymbol{p}_j^t$, is uniform among N cells. In HIST algorithm, clients and edge servers exchange model information every H iterations. Consequently, the average per-iteration complexity of each client for uploading the local model can be expressed as $\frac{L_0}{NH}$. Similarly, for each edge server, the average complexity per iteration for disseminating the edge model is $\frac{n_j L_0}{NH}$. In addition to the edge model dissemination, edge servers also upload their models to the cloud server at intervals of every HE iterations. This process incurs an extra transmission load of $\frac{L_0}{N}$. As a result, the total communication complexity for each edge server is $\frac{n_j L_0}{NH} + \frac{L_0}{NHE}$ per iteration. Additionally, the complexity of the global model dissemination can be quantified as $\frac{L_0}{HE}$ for the global server. In summary, submodel partitioning in HIST effectively reduces the communication consumption for the global server, edge servers, and client to $\frac{1}{N}$ of that required by the conventional hierarchical FedAvg.

III. Convergence Analysis

This section provides convergence analysis for the proposed HIST algorithm. We note that the convergence proof of conventional hierarchical FedAvg cannot be directly extended to our case that adopts *submodel partitioning*, due to the effect of the masks. Specifically, when comparing $\boldsymbol{p}_j^t \odot \nabla F_i(\boldsymbol{p}_j^t \odot \boldsymbol{x})$ and $\nabla F_i(\boldsymbol{x})$, the mask $\boldsymbol{p}_j^t$ compresses not only the gradient but also the model, while many existing works only investigate compressing the gradient. Theoretical analysis on model compression [31] is quite limited. Even in the single-cell scenario, existing proofs of IST [22], [23] rely on some stronger assumptions. The hierarchical architecture with both multiple steps of client update and multiple steps of edge training we consider makes the analysis further complicated.

A. Assumptions

We focus on a general non-convex loss function and consider a non-i.i.d data setting. Our theoretical analysis relies on the following assumptions.

Assumption 1. *The global loss function $f(\boldsymbol{x})$ has a lower bound f_*, i.e., $f(\boldsymbol{x}) \geq f_*, \forall \boldsymbol{x}$.*

Assumption 2. *F_i is differentiable and L-smooth, i.e., there exists a positive constant L such that for any $\boldsymbol{x}$ and $\boldsymbol{y}$*

$$\begin{aligned} &\|\nabla F_i(\boldsymbol{x}) - \nabla F_i(\boldsymbol{y})\|^2 \leq L\|\boldsymbol{y} - \boldsymbol{x}\|, \forall i, \\ &F_i(\boldsymbol{y}) \leq F_i(\boldsymbol{x}) + \langle \nabla F_i(\boldsymbol{x}), \boldsymbol{y} - \boldsymbol{x} \rangle + \frac{L}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2, \forall i. \end{aligned} \quad (9)$$

With Assumption 2, one can also claim that functions f_j, $\forall j$ and f are L-smooth [32].

Assumption 3. *The stochastic gradient $\nabla l(\boldsymbol{x}, \xi_i)$ is an unbiased estimator of the true gradient, i.e., $\mathbb{E}_{\xi_i \sim \mathcal{D}_i}[\nabla l(\boldsymbol{x}, \xi_i)] = \nabla F_i(\boldsymbol{x}), \forall \boldsymbol{x}, \forall i$.*

Assumption 4. *The variance of the stochastic gradient $\nabla l(\boldsymbol{x}, \xi_i)$ is bounded as*

$$\mathbb{E}_{\boldsymbol{\xi}_i \sim \mathcal{D}_i} \|\nabla l(\boldsymbol{x}, \xi_i) - \nabla F_i(\boldsymbol{x})\|^2 \leq \sigma^2, \forall \boldsymbol{x}, \forall i. \tag{10}$$

Assumption 5. *There exists a constant δ_1^2 that bounds the gradient dissimilarity between the global loss function and edge loss functions, i,e.,*

$$\frac{1}{N} \sum_{j=1}^{N} \|\nabla f_j(\boldsymbol{x}) - \nabla f(\boldsymbol{x})\|^2 \leq \delta_1^2, \forall \boldsymbol{x}. \tag{11}$$

Assumption 6. *There exists a constant δ_2^2 that bounds gradient dissimilarities between edge loss functions and client loss functions, i,e.,*

$$\frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \|\nabla F_i(\boldsymbol{x}) - \nabla f_j(\boldsymbol{x})\|^2 \leq \delta_2^2, \forall \boldsymbol{x}, \forall j. \tag{12}$$

Assumptions 1-4 have been widely adopted in the context of stochastic non-convex and smooth optimization [32]–[34]. Assumptions 5 and 6 serve to characterize the degree of data heterogeneity across cells and across clients, which are commonly used within the HFL literature [9], [35], [36].

B. Theoretical Results

The global synchronization (7) occurs at intervals of every E step of edge model updates. To capture the evolution of the global model better, we introduce iterates $\{\hat{\boldsymbol{x}}^{t,e} = \sum_{j=1}^{N} \bar{\boldsymbol{x}}_j^{t,e} \mid t = 0, 1, \ldots, T-1; e = 0, 1, \ldots, E-1\}$ serving as a virtual sequence of global models, which is commonly employed to monitor the convergence of distributed algorithms with delayed global synchronization [37].

We investigate the convergence properties of the proposed algorithm by characterizing the evolution of $\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\|^2$ to capture how fast the global model converges to a stationary point of a general non-convex loss function. Now we state the following main theorem.

Theorem 1. *Suppose that Assumptions 1-6 hold, $N \geq 2$, and the step size satisfies*

$$\gamma \leq \min\left\{\frac{1}{45\sqrt{N}EHL}, \frac{\tilde{N}}{NHL}, \frac{1}{NH^2L}, \frac{1}{N(N+1)E^2H^2L}\right\}. \tag{13}$$

Then the HIST *algorithm satisfies*

$$\begin{aligned}\frac{1}{TE}\sum_{t=0}^{T}\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 &\leq 4\frac{f(\bar{\boldsymbol{x}}^0) - f_*}{\gamma TEH} + 100\gamma\tilde{N}L\sigma^2 \\ &+ 1356\gamma L\delta_1^2 + 60\gamma L\delta_2^2 + 6\frac{N}{d}\frac{1}{T}\sum_{t=0}^{T-1} d_{max}^t \delta_1^2 \\ &+ 48(N-1)L^2\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\bar{\boldsymbol{x}}^t\right\|^2,\end{aligned} \tag{14}$$

where $\tilde{N} = \sum_{j=1}^{N} \frac{1}{n_j}$ and

$$d_{max}^t = \max\{\|\boldsymbol{p}_1^t\|_1, \|\boldsymbol{p}_2^t\|_1, \ldots, \|\boldsymbol{p}_N^t\|_1\}, \forall t. \tag{15}$$

Proof. The proof is provided in Appendix A □

Theorem 1 presents an upper bound for the optimality gap of HIST characterized by the time-averaged squared gradient norm of the global virtual model sequence. The first term in this upper bound exhibits the effect of the initial optimality gap $f(\bar{\boldsymbol{x}}^0) - f_*$ on the convergence behavior of HIST. The second term shows how the optimality gap is related to the variance of stochastic gradients σ^2. This variance can be mitigated by enlarging the mini-batch size during the computation of stochastic gradients, e.g., replacing SGD (i.e., mini-batch size of 1) with a mini-batch SGD. The third, fourth, and fifth terms reflect the influence of non-i.i.d. characteristics within the cell (δ_2^2) and across cells (δ_1^2) on convergence. The last term demonstrates that the norm of the synchronized global model $\|\bar{\boldsymbol{x}}^t\|^2$ also impacts the optimality gap. Note that the last two terms are induced by submodel partitioning. In addition, the step size γ is a configurable parameter that affects the first four terms of the derived upper bound. Plugging an appropriate step size into Theorem 1 gives rise to the following corollary.

Corollary 1. *Suppose that Assumptions 1-6 hold, and the masks $\{\boldsymbol{p}_1^t, \boldsymbol{p}_2^t, \ldots, \boldsymbol{p}_N^t\}$ are uniformly and randomly generated based on (3). Let the step size $\gamma = (TEH\tilde{N})^{-\frac{1}{2}}$ in which T is large enough to satisfy (13). Then for $N \geq 2$, the* HIST *algorithm satisfies*

$$\begin{aligned}\frac{1}{TE}\sum_{t=0}^{T}\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 &\leq \mathcal{O}\left(\tilde{N}^{\frac{1}{2}}T^{-\frac{1}{2}}\right) + \mathcal{O}\left(T^{-\frac{1}{2}}\right) \\ &+ 6\delta_1^2 + 48(N-1)L^2\frac{1}{T}\sum_{t=0}^{T-1}\left\|\bar{\boldsymbol{x}}^t\right\|^2,\end{aligned} \tag{16}$$

where $\tilde{N}$ is described in Theorem 1.

Remark 1. *In Corollary 1, the first term is a function of $\tilde{N} = \sum_{j=1}^{N} \frac{1}{n_j}$, which is affected by the relationship between the number of clients in each cell, denoted as n_j, and the total number of cells, denoted as N. When the number of clients in each cell is of a comparable magnitude or greater than the total number of cells, the convergence rate of the diminishing terms in the derived upper bound is primarily determined by $\mathcal{O}\left(T^{-\frac{1}{2}}\right)$. On the other hand, if the number of clients in each cell is significantly smaller in relation to the total number of cells, $\tilde{N}$ becomes influential, and the convergence rate is dominated by $\mathcal{O}\left(\tilde{N}^{\frac{1}{2}}T^{-\frac{1}{2}}\right)$.*

C. Discussions

Non-diminishing bound: With the step size chosen in Corollary 1, the first four terms in (14) will diminish to zero as long as the number of total iterations T is large enough. The rest two terms are non-diminishing parts that arise due to submodel training. One can claim that HIST can converge to the neighborhood of a stationary point of the loss function under the aforementioned conditions. A similar phenomenon has also been reported in the single-cell case [18], [24]. The

bound enables us to explore the performance-resource trade-off, where more detailed discussions will be provided in the remainder of this subsection.

The choice of N: As N increases, i.e., as the overall clients in the system are divided into more cells during training, the size of the submodels gets smaller, providing a more lightweight model to edge servers and clients. As a result, the training costs including computation, communication, and storage will be reduced at each iteration. However, as observed in Corollary 1, a large N causes the sequence to deviate further from the stationary point. Overall, there is a trade-off between the convergence performance and computation, communication, and storage costs.

The effect of n_j: Under bounded heterogeneity assumptions across cells and clients, i.e., under Assumptions 5 and 6, the number of clients n_j in each cell j has a positive effect on the convergence. To be specific, based on the expression of $\tilde{N}$, one can claim that $\tilde{N}$ monotonically decreases as n_j increases for all j. Meanwhile, according to Corollary 1, the smaller the value of $\tilde{N}$, the faster the convergence.

The choices of H and E: The choices of H and E impact the communication frequency. As H increases, the aggregation frequency at edge servers will become smaller, reducing the communication load between clients and the edge server. On the other hand, a large E induces fewer global synchronizations, which alleviates the communication burden between edge servers and the cloud server. However, these values cannot become infinitely large. The maximum values of H and E can be derived from the condition of the step size γ. Specifically, to ensure that the step size $\gamma = (TEH\tilde{N})^{-\frac{1}{2}}$ adheres to the conditions specified in (13), H and E can be set as on the order of $\mathcal{O}\left((\tilde{N}T)^{\frac{1}{3}}N^{-\frac{4}{3}}H^{-1}\right)$ and $\min\left\{\mathcal{O}\left((\tilde{N})^3TEN^2\right), \mathcal{O}\left((\tilde{N}T)^{\frac{1}{3}}N^{-\frac{4}{3}}E^{-1}\right)\right\}$ at most, respectively.

The effect of submodel partitioning: According to equations (14) and (15) in Theorem 1, a uniform partition with $\|\boldsymbol{p}_1^t\|_1 = \ldots = \|\boldsymbol{p}_N^t\|_1 = d/N$ leads the second to last term of the upper bound i.e., $6\frac{N}{d}\frac{1}{T}\sum_{t=0}^{T-1} d_{\max}^t \delta_1^2$, to attain its minimum. Nevertheless, this strategy may not be the optimal in terms of latency and resource efficiency when the communication and computation resources are heterogeneous across the clients. There is a trade-off between learning performance and training efficiency depending on the choice of mask sizes. This motivates us to optimize the mask sizes to balance the training efficiency and the convergence bound in the next section.

D. Comarison with Convergence Analysis of Existing Works

The key distinction between ours and previous works [18], [22]–[24] on IST is that we consider the hierarchical architecture in this paper. Apart from this, our focus and assumptions are also different from these works. To be specific, the analysis of [18], [22] concentrates on the centralized setting. Consequently, the non-i.i.d. influence is not encompassed in their works, which is an indispensable factor in distributed training. Additionally, [18], [22], [23] make use of relatively stronger assumptions in their analysis. In particular, [18], [23] assume Lipschitz continuity for the loss function. Moreover, Assumption 5 adopted in [23] is also not general. On the other hand, [22] imposes some extra conditions on the masks, which may not hold in practical settings as shown in [24]. The authors in [24] provide a tighter convergence bound for distributed IST based on mild assumptions. Nevertheless, their analysis focuses on the quadratic function, which is challenging to be extended to general smooth functions.

IV. Optimization of Submodel Partitioning

In this section, we propose to optimize submodel partitioning sizes to balance training efficiency and convergence bound.

A. Training Latency Analysis

Since the edge and cloud servers possess significantly larger communication and computation resources than clients, we focus on the delays for local model updates and uplink communication at the clients, as done in [38]–[40]. Additionally, we assume that the computational and communication capabilities of clients remain stable throughout a global training round [28], [41]. Specifically, we use $\mathcal{R}_i^t$ and $\mathcal{F}_i^t$ to represent the data rate for uplink communication and CPU frequency of client i in t-th global training round, respectively. Furthermore, the communication load for uploading a complete model is indicated by L_0, while the required number of CPU cycles for a single mini-batch update of the full model is represented by V_0. For analytical tractability, we assume that the latency required for local computation is linearly proportional to the model size[1] following [42].

Based upon the traditional time division multiple access (TDMA) transmission[2], which is one of the standard OMA protocols, the overall latency for the (t, e)-th round within cell j can be expressed as

$$\max_{i\in\mathcal{C}_j}\left\{H\frac{\|\boldsymbol{p}_j^t\|_1 V_0}{\mathcal{F}_i^t d} + n_j\frac{\|\boldsymbol{p}_j^t\|_1 L_0}{\mathcal{R}_i^t d}\right\}. \tag{17}$$

Given that the duration required for a single update of the global model is determined by the speed of the slowest edge server, we can express the overall latency for each global model update as follows:

$$\max_{j\in\{1,2,\ldots,N\}}\left\{E\max_{i\in\mathcal{C}_j}\left\{H\frac{\|\boldsymbol{p}_j^t\|_1 V_0}{\mathcal{F}_i^t d} + n_j\frac{\|\boldsymbol{p}_j^t\|_1 L_0}{\mathcal{R}_i^t d}\right\}\right\}. \tag{18}$$

In the following, we will show how we can optimize the mask sizes $\{\|\boldsymbol{p}_1^t\|_1, \|\boldsymbol{p}_2^t\|_1, \ldots, \|\boldsymbol{p}_N^t\|_1\}$ for individual cells, which affect not only the overall training latency in (18) but also the convergence bound in (14).

B. Mask Size Optimization

Note that the mask sizes affect both the latency and the convergence bound. In particular, Theorem 1 shows that an imbalanced mask size distribution results in a large deviation

[1]The analysis can be easily extended to any other functions that describe the relationship between the model size and the computation time.

[2]Extending the analysis to the orthogonal frequency division multiple access (OFDMA) case is straightforward.

from the stationary point for the HIST algorithm. Moreover, the uniform mask partition leads to the minimum convergence bound, which however does not lead to the minimum latency. Choosing the size of the mask involves a compromise between the efficiency of training and the performance of convergence. To suppress the convergence bound induced by imbalanced mask size distribution, we enforce the second to last item of the convergence bound in (14) below to a predefined threshold ϵ_{th}. The latency minimization problem of HIST is thus formulated as

$$\min_{\{\|\boldsymbol{p}_j^t\|_1\}_{j=1}^N} \max_{j\in\{1,\ldots,N\}} \left\{ E \max_{i\in\mathcal{C}_j} \left\{ H\frac{\|\boldsymbol{p}_j^t\|_1 V_0}{\mathcal{F}_i^t d} + n_j \frac{\|\boldsymbol{p}_j^t\|_1 L_0}{\mathcal{R}_i^t d} \right\} \right\} \tag{19a}$$

$$\text{s.t.} \quad \sum_{j=1}^N \|\boldsymbol{p}_j^t\|_1 = d \tag{19b}$$

$$6\frac{N\|\boldsymbol{p}_j^t\|_1}{d}\delta_1^2 \le \epsilon_{\text{th}},\ j=1,2,\ldots,N, \tag{19c}$$

where ϵ_{th} denotes a preset threshold to guarantee a certain level of learning performance. From the perspective of the optimal solution, problem (19) is equivalent to the following mixed integer linear programming problem:

$$\begin{aligned} \min_{\{\|\boldsymbol{p}_j^t\|_1\}_{j=1}^N} \quad & t \\ \text{s.t.} \quad & \max_{i\in\mathcal{C}_j} \left\{ H\frac{\|\boldsymbol{p}_j^t\|_1 V_0}{\mathcal{F}_i^t d} + n_j \frac{\|\boldsymbol{p}_j^t\|_1 L_0}{\mathcal{R}_i^t d} \right\} \le t,\ \forall j \\ & \sum_{j=1}^N \|\boldsymbol{p}_j^t\|_1 = d \\ & 6\frac{N\|\boldsymbol{p}_j^t\|_1}{d}\delta_1^2 \le \epsilon_{\text{th}},\ j=1,2,\ldots,N, \end{aligned} \tag{20}$$

which can be tackled by standard optimization solvers, e.g., Gurobi. At the commencement of each global training round, the cloud server just needs to randomly generate masks based on these optimized mask sizes, partition the current global model accordingly, and send the obtained submodel partitions to the corresponding cells.

V. AirComp-Assisted HIST Algorithm

In this section, we propose an AirComp version of HIST, which takes advantage of the superposition property of multiple access channels. We consider a hierarchical network where edge servers communicate with clients over wireless channels, while the cloud server exchanges model information with the edge servers by wired links.

Note that through our submodel partitioning strategy in HIST, the per-iteration communication complexity at the cloud server remains consistent regardless of the number of edge servers. Conversely, the communication complexity experienced by edge servers increases in proportion to their client counts. Owing to the limited availability of radio resources, the communication bottleneck at edge servers becomes increasingly pronounced in the context of wireless networks. Motivated by this, we incorporate AirComp into the HIST methodology, aiming to facilitate efficient model aggregation at each edge server. This ensures that the model aggregation delay at each edge server becomes independent of the number of clients within its coverage, thus alleviating the communication burden at the edge servers. We assume that the downlink model dissemination in each cell is error-free, as edge servers possess sufficiently large transmit power compared to resource-constrained clients [38]–[40]. Our primary focus will be on the uplink model uploading within each cell.

Different from the existing works on AirComp-assisted FL, each edge server is in charge of aggregating a different part of the model compared to other edge servers. There is no superposition effect in terms of error from different edge servers induced by AirComp. Additionally, the communication load of model aggregations in each edge server is configurable. In this section, we will propose to optimize the submodel partitioning sizes by minimizing the training latency of the AirComp-assisted HIST algorithm.

A. Signal Transmission Model

In Step 7 of Algorithm 1, each edge server aims to acquire an average of local models of clients within its coverage area, as represented by (6). We adopt AirComp to support this aggregation, which allows each edge server to directly estimate an average of signals transmitted from their clients, bypassing the decoding of individual signals. To mitigate the effect of wireless noise, instead of aggregating models, we let each client i in cell j upload the accumulated gradient $\boldsymbol{\delta}_i^{t,e}$ during H-step local updates at the end of (t,e)-th round of edge training. Specifically, $\boldsymbol{\delta}_i^{t,e}$ can be written as

$$\boldsymbol{\delta}_i^{t,e} = \frac{1}{\gamma}\left(\boldsymbol{x}_{i,H}^{t,e} - \boldsymbol{x}_{i,0}^{t,e}\right) = \sum_{h=0}^{H-1} \boldsymbol{p}_j^t \odot \nabla l(\boldsymbol{x}_{i,h}^{t,e}, \xi_{i,h}^{t,e}). \tag{21}$$

Under this scheme, the impact of wireless noise on the aggregated model will be mitigated by the ratio γ compared to directly uploading the models [41], [43], as we will see later.

Consider a single-input-multiple-output (SIMO) AirComp system, where clients are deployed with a single antenna and edge servers have M antennas. Without loss of generality, we focus on a specific time slot where each client i in cell j concurrently uploads the k-th element of $\boldsymbol{\delta}_i^{t,e}$, i.e., $(\delta_i^{t,e})_k$, to the j-th edger server during the (t,e)-th edge round. The goal of the edge server is to obtain $(\bar{\delta}_j^{t,e})_k := \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}(\delta_i^{t,e})_k$, while the received signal becomes

$$(\hat{\delta}_j^{t,e})_k = (\boldsymbol{m}_j^{t,e})^{\text{H}} \left(\sum_{i\in\mathcal{C}_j} \boldsymbol{h}_i^{t,e} \alpha_i^{t,e} (\delta_i^{t,e})_k + \boldsymbol{z}_{j,k}^{t,e} \right), \tag{22}$$

where $\alpha_i^{t,e}$ denotes the precoding factor at client i and $\boldsymbol{h}_i^{t,e} \in \mathbb{C}^M$ represents the SIMO channel between client i and edge server j which is assumed to be invariant during (t,e)-th communication round. We assume that $\boldsymbol{h}_i^{t,e}$ is known to clients and edge servers as in [38], [43]. $\boldsymbol{m}_j^{t,e} \in \mathbb{C}^M$ denotes the receive beamforming vector at edge server j, and $\boldsymbol{z}_{j,k}^{t,e} \sim \mathcal{CN}(0, \sigma_0^2 \boldsymbol{I}_M)$ represents the additive white Gaussian noise (AWGN). The precoding factor of clients in cell j is constrained by the maximum power, i.e., $\|\alpha_i^{t,e}\| \le P_j$.

B. Aggregation Distortion of AirComp

The objective of AirComp is to acquire an estimate of $(\bar{\delta}_j^{t,e})_k$. It can be observed from (22) that the distortion between the received signal and the target model average comes from the misalignment of fading channels and the noise. To mitigate the effect of non-uniform fading channels, we set the precoding factor $\alpha_i^{t,e}$ as

$$(\alpha_i^{t,e})^\star = \frac{1}{n_j}\frac{\left((\boldsymbol{m}_j^{t,e})^{\mathrm{H}}\boldsymbol{h}_i^{t,e}\right)^\dagger}{\|(\boldsymbol{m}_j^{t,e})^{\mathrm{H}}\boldsymbol{h}_i^{t,e}\|^2},\ \forall\, i\in\mathcal{C}_j. \tag{23}$$

To meet the energy constraint $\|\alpha_i^{t,e}\|^2 \le P_j$, let $\boldsymbol{m}_j^{t,e} = \frac{1}{\nu_j^{t,e}}\boldsymbol{a}_j^{t,e}$ where $\nu_j^{t,e} = \sqrt{P}\min_{i\in\mathcal{C}_j}\|(\boldsymbol{a}_j^{t,e})^{\mathrm{H}}\boldsymbol{h}_i^{t,e}\|$ and $\|\boldsymbol{a}_j^{t,e}\| = 1$, in which $\nu_j^{t,e}$ denotes the power normalization factor and $\boldsymbol{a}_j^{t,e}$ represents the normalized receive beamformer.

Under the precoding factor (23), the signal shown in (22) is thus simplified to

$$(\hat{\delta}_j^{t,e})_k = \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}(\delta_i^{t,e})_k + \frac{1}{\nu_j^{t,e}}(\boldsymbol{a}_j^{t,e})^{\mathrm{H}}\boldsymbol{z}_{j,k}^{t,e}, \tag{24}$$

which is an unbiased estimator of $(\bar{\delta}_j^{t,e})_k$. The distortion of AirComp measured by mean-squared error (MSE), also known as the variance of $(\hat{\delta}_j^{t,e})_k$, is thus given by

$$\mathrm{MSE}_{j,k}^{t,e} = \frac{\sigma_0^2}{P\min_{i\in\mathcal{C}_j}\|(\boldsymbol{a}_j^{t,e})^{\mathrm{H}}\boldsymbol{h}_i^{t,e}\|^2}. \tag{25}$$

Consequently, the whole local model aggregation at (t,e)-th round in cell j admits the following expression:

$$\hat{\boldsymbol{\delta}}_j^{t,e} = \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{\delta}_i^{t,e} + \boldsymbol{Z}_j^{t,e}, \tag{26}$$

where $\boldsymbol{Z}_j^{t,e}$ denotes a noise vector and its k-th element is $1/\nu_j^{t,e}(\boldsymbol{a}_j^{t,e})^{\mathrm{H}}\boldsymbol{z}_{j,k}^{t,e}$ if $(p_j^t)_k \neq 0$, and 0, otherwise. It is unnecessary to resort to AirComp for aggregation if $(p_j^t)_k = 0$. In addition, all the edge servers fulfill this aggregation in parallel.

Now each edge server j updates the edge model as

$$\bar{\boldsymbol{x}}_j^{t,e+1} = \bar{\boldsymbol{x}}_j^{t,e} - \gamma\hat{\boldsymbol{\delta}}_j^{t,e}, \forall j, \tag{27}$$

which is an unbiased estimator of the average of local models in $\mathcal{C}_j$, exhibiting a variance quantized by

$$\gamma^2\mathbb{E}\|\boldsymbol{Z}_j^{t,e}\|^2 = \gamma^2\sum_{k\in\mathcal{S}_j^t}\mathrm{MSE}_{j,k}^{t,e},\ \mathcal{S}_j^t = \{k|(p_j^t)_k \neq 0\}. \tag{28}$$

With the combination of HIST and AirComp, the communication complexity at each edge server is independent of the number of clients within the coverage.

Remark 2. *By aggregating the accumulated gradient and then updating the edge model via a step SGD at as shown in (27), the impact of channel noise on the convergence of the AirComp-Assisted HIST algorithm can be analogized to the noise induced by the inherent randomness of stochastic gradients. Such a strategy facilitates the convergence analysis and allows us to select an appropriate step size γ to mitigate the effect of channel noise.*

C. Convergence Analysis of AirComp-assisted HIST

Next, we analyze the convergence behavior of the AirComp-assisted HIST algorithm. The result is provided in Theorem 2 below.

Theorem 2. *Suppose that Assumptions 1-6 hold, $N \ge 2$, and the step size satisfies condition (13). Then the AirComp-assisted HIST algorithm satisfies*

$$\begin{aligned}\frac{1}{TE}\sum_{t=0}^{T}\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 &\le 4\frac{f(\bar{\boldsymbol{x}}^0)-f_*}{\gamma TEH} + 100\gamma\tilde{N}L\sigma^2\\ &+ 1356\gamma L\delta_1^2 + 60\gamma L\delta_2^2 + 48(N-1)L^2\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left\|\bar{\boldsymbol{x}}^t\right\|^2\\ &+ \gamma\frac{1}{TE}\sum_{t=0}^{T}\sum_{e=0}^{E-1}\sum_{j=1}^{N}\mathrm{MSE}_j^{t,e} + 6\frac{N}{d}\frac{1}{T}\sum_{t=0}^{T-1}d_{max}^t\delta_1^2,\end{aligned} \tag{29}$$

where $\mathrm{MSE}_j^{t,e} = \sum_{k\in\mathcal{S}_j^t}\mathrm{MSE}_{j,k}^{t,e}$, $\mathcal{S}_j^t = \{k|p_j^t \neq 0\}$, and $\tilde{N}$ and d_{max}^t are defined in Theorem 1.

Proof. The proof is provided in Appendix A □

Plugging an appropriate step size into Theorem 2 gives rise to the following corollary.

Corollary 2. *Suppose that Assumptions 1-6 hold and let the step size $\gamma = (TEH\tilde{N})^{-\frac{1}{2}}$ in which T is large enough to satisfy (13). Then the HIST algorithm satisfies*

$$\begin{aligned}\frac{1}{TE}\sum_{t=0}^{T}\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 &\le \mathcal{O}\left(\tilde{N}^{\frac{1}{2}}T^{-\frac{1}{2}}\right) + \mathcal{O}\left(T^{-\frac{1}{2}}\right)\\ &\underbrace{+\gamma\frac{1}{TE}\sum_{t=0}^{T}\sum_{e=0}^{E-1}\sum_{j=1}^{N}\mathrm{MSE}_j^{t,e} + 6\frac{N}{d}\frac{1}{T}\sum_{t=0}^{T-1}d_{max}^t\delta_1^2}_{\text{controlable terms}}\\ &+ 48(N-1)L^2\frac{E}{T}\sum_{m=0}^{T/E-1}\mathbb{E}\left\|\bar{\boldsymbol{x}}^t\right\|^2.\end{aligned} \tag{30}$$

As shown in Theorem 2, the error term induced by AirComp is of the same order as the variance of the stochastic gradient. This similarity is attributed to the fact that, under the precoding design as specified in (23), the accumulated gradient estimated by AirComp remains unbiased. Consequently, the variance affects the algorithm in a manner akin to the inherent randomness of the stochastic gradient and just adds more uncertainty. It thus slows the convergence of HIST. The next subsection will focus on minimizing AirComp distortion by designing the receiver beamforming vectors.

D. Receive Beamforming Design

This subsection aims to design the normalized receive beamformers to mitigate the impact of AirComp distortion, by minimizing the MSE term in Theorem 2. Note that the expression of MSE in (25) is invariant to k, i.e., $\mathrm{MSE}_{j,k}^{t,e} = \mathrm{MSE}_{j,k'}^{t,e}$,

$\forall k, k' \in \mathcal{S}_j^t$. Therefore, the beamformer design problem at the (t,e)-th round can be formulated as

$$\min_{\boldsymbol{a}_1^{t,e},\boldsymbol{a}_2^{t,e},\ldots,\boldsymbol{a}_N^{t,e}} \sum_{i=1}^{N} \frac{|\mathcal{S}_j^t|\sigma_0^2}{P\min_{i\in\mathcal{C}_j}\|(\boldsymbol{a}_j^{t,e})^{\mathrm{H}}\boldsymbol{h}_i^{t,e}\|^2} \quad \text{s.t.} \quad \|\boldsymbol{a}_j^{t,e}\|^2 = 1,\ j=1,2,\ldots,N, \tag{31}$$

where $|\mathcal{S}_j^t|$ denotes the cardinality of set $\mathcal{S}_j^t$.

Due to the independence between $\boldsymbol{a}_j^{t,e}$ and $\boldsymbol{a}_{j'}^{t,e}$, $\forall j \neq j'$, problem (31) can be transformed into N independent subproblems as

$$\begin{aligned} &\max_{\boldsymbol{a}_j^{t,e}} \min_{i\in\mathcal{C}_j} \|(\boldsymbol{a}_j^{t,e})^{\mathrm{H}}\boldsymbol{h}_i^{t,e}\|^2 \\ &\text{s.t. } \|\boldsymbol{a}_j^{t,e}\|^2 = 1,\ j=1,2,\ldots,N. \end{aligned} \tag{32}$$

While $\boldsymbol{a}_j^{t,e}$ is subject to a unimodular constraint, problem (32) can be directly relaxed to

$$\begin{aligned} &\max_{\boldsymbol{a}_j^{t,e}} \min_{i\in\mathcal{C}_j} \|(\boldsymbol{a}_j^{t,e})^{\mathrm{H}}\boldsymbol{h}_i^{t,e}\|^2 \\ &\text{s.t. } \|\boldsymbol{a}_j^{t,e}\|^2 \leq 1,\ j=1,2,\ldots,N, \end{aligned} \tag{33}$$

without compromising optimality. The reason is that the optimal solution $(\boldsymbol{a}_j^{t,e})^*$ of problem (33) must be in the boundary of its domain. If it were not, projecting it onto the boundary with $(\boldsymbol{a}_j^{t,e})^\dagger = (\boldsymbol{a}_j^{t,e})^*/\|(\boldsymbol{a}_j^{t,e})^*\|_2$ would yield a higher objective function value, thereby contradicting the optimality of $(\boldsymbol{a}_j^{t,e})^*$. Although (33) is a non-convex quadratically constrained quadratic programming (QCQP) problem, it can be approximately solved using a highly efficient first-order method, namely, the Mirror-Prox-based Successive Convex Approximation (SCA) algorithm, as detailed in [13], [44].

E. Submodel Partitioning Optimization

When AirComp is incorporated into our framework, the uplink communication delay at cell j can be written according to [39], [45] as

$$\frac{\|\boldsymbol{p}_j^t\|_1}{B/\Delta f} t_s, \tag{34}$$

where B denotes the system bandwidth and Δf denotes the bandwidth of a OFDM subcarrier, and t_s denotes the duration of an OFDM symbol. In this work, we follow 5G New Radio (NR) protocol and set $\Delta f = 15$kHZ and $t_s = \frac{1}{14}$ ms [46]. Additionally, the computation latency here is the same as in the first term of (17). The total latency for each edge training round within cell j can be represented as

$$\max_{i\in\mathcal{C}_j} \left\{ H\frac{\|\boldsymbol{p}_j^t\|_1 V_0}{\mathcal{F}_i^t d} + \frac{\|\boldsymbol{p}_j^t\|_1}{B/\Delta f} t_s \right\}. \tag{35}$$

As a result, the latency minimization problem of this AirComp-assisted HFL system can be formulated as

$$\min_{\{\|\boldsymbol{p}_j^t\|_1\}_{j=1}^N} \max_{j\in\{1,\ldots,N\}} \left\{ E \max_{i\in\mathcal{C}_j} \left\{ H\frac{\|\boldsymbol{p}_j^t\|_1 V_0}{\mathcal{F}_i^t d} + \frac{\|\boldsymbol{p}_j^t\|_1}{B/\Delta f} t_s \right\} \right\} \quad \text{s.t.} \quad (19b), (19c), \tag{36}$$

which can be resolved using a method similar to that employed for problem (19). The implementation procedure of the AirComp-assisted HIST algorithm is illustrated in Fig. 2.

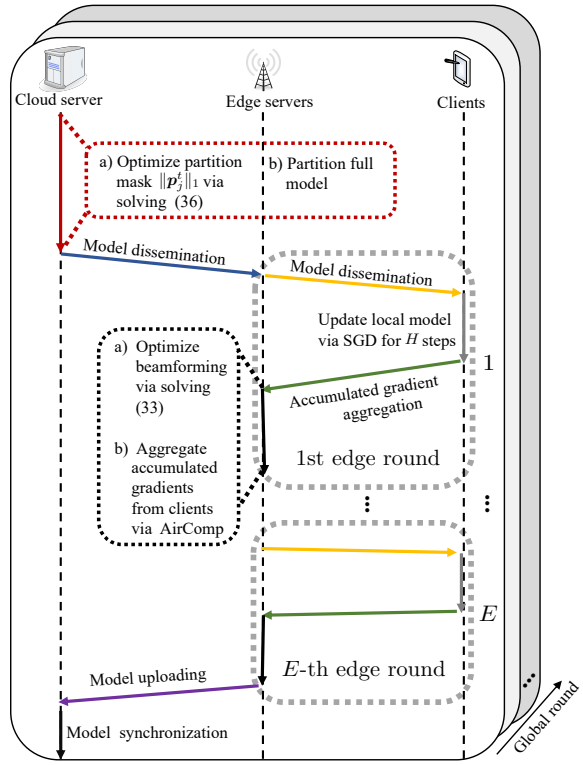


Fig. 2: The overall implementation of the AirComp-assisted HIST algorithm.

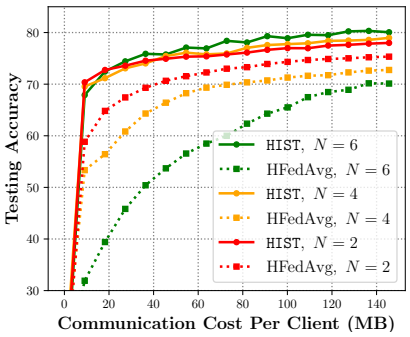
VI. Simulations

In this section, we conduct numerical experiments to evaluate the performance of the proposed HIST algorithm. The code of our methodology is available at https://wenzhifang.github.io/.

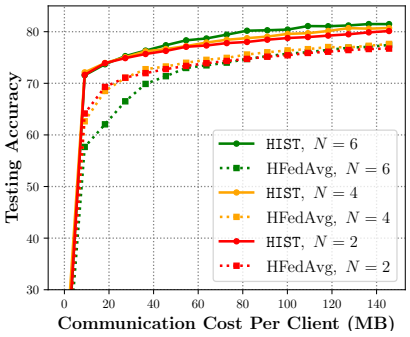
A. Simulation Settings

We consider a setup in which clients are evenly distributed across N cells, i.e., $n_j = n_{j'}, \forall j, j' \in \{1,2,\ldots,N\}$. We consider various scenarios with $N \in \{2,4,6,8\}$. We focus on two practical data distribution settings: (i) the fully non-i.i.d. case and (ii) the case with i.i.d. data across cells but non-i.i.d data across the clients within the same cell. For the former case, the client's dataset construction follows the approach outlined in [33]. The overall training set is first sorted based on their corresponding labels. The sorted training set is then partitioned into $2Nn_j$ shards, with each shard containing the same number of samples as others. Subsequently, each client is assigned 2 shards uniformly at random, ensuring all the data samples are evenly distributed to clients. For the latter case, we first uniformly and randomly divide the entire training set into N parts, corresponding to N cells, and then distribute each part to the clients within the respective cell in a non-i.i.d. manner following the former case. Unless otherwise specified, the total number of clients, i.e., Nn_j, in the system is set to 60.

We train a fully connected neural network and a convolutional neural network using Fashion-MNIST and CIFAR-10 datasets, where the details are depicted in the following subections.

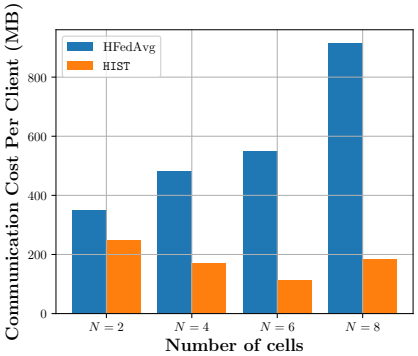


(a) Fully non-i.i.d.

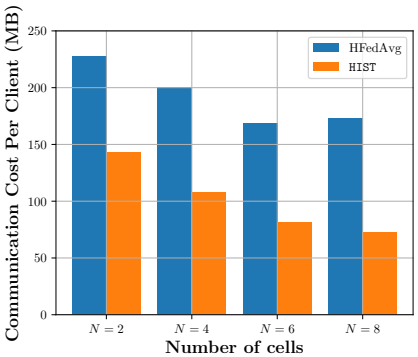


(b) Cell i.i.d., client non-i.i.d.

Fig. 3: The impact of the number of cells N on the convergence performance.

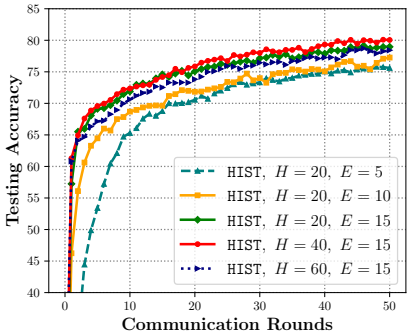


(a) Fully non-i.i.d.

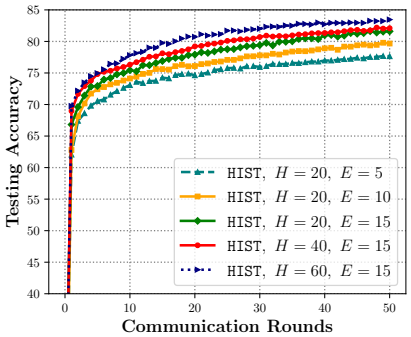


(b) Cell i.i.d., client non-i.i.d.

Fig. 4: Comparison of communication cost for achieving the testing accuracy of 80%.



(a) Fully non-i.i.d.



(b) Cell i.i.d., client non-i.i.d.

Fig. 5: The impacts of aggregation periods H and E on the convergence performance.

B. Performance Evaluation of `HIST`

1) Fully connected neural network: We consider an image classification task on Fashion-MNIST using a two-layer fully connected neural network. In this model, we configure the input layer to have 784 neurons, corresponding to the size of the input image, and the output layer to have 10 neurons, which matches the number of classes. Additionally, we employ a hidden layer with 300 neurons. The size of this model is 0.91 MB. The cloud server disjointly partitions hidden neurons to construct different submodels, which is the implementation strategy of masks defined in (3) and [18], [23]. In this subsection, we first consider the case of the uniform partition, i.e., all submodels have the same sizes. This can be achieved by uniformly and randomly partitioning the hidden neurons. The difference between our implementation and [18], [23] are discussed in Appendix B.

Comparison with Baselines: In Fig. 3, we compare our proposed `HIST` algorithm with the traditional hierarchical FedAvg (denoted as HFedAvg in our figures) where the full model is communicated over the hierarchical network. We evaluate the performance by comparing testing accuracy across different numbers of cells, $N \in \{2, 4, 6\}$. The X-axis here represents the communication load which quantifies the volume of parameters transmitted per client. We set H and E to 20 and 5, respectively. As shown in Figs. 3a and 3b, the proposed `HIST` outperforms HFedAvg in terms of testing accuracy at the same levels of communication cost for both data distribution settings. In particular, under a fully non-i.i.d. data setup, as N increases, the non-i.i.d. extent of data among cells becomes more pronounced, leading to performance degradation for HFedAvg. In contrast, `HIST` achieves a higher testing accuracy when N increases from 2 to 6. This is because, for `HIST`, the per-round communication cost per client decreases as the number of cells increases, partly compensating for the degradation of convergence induced by data heterogeneity.

Fig. 4 compares the communication cost of `HIST` and HFedAvg for achieving a desired accuracy. This experiment was also carried out with $H = 20$ and $E = 5$. The desired testing accuracy is set to 80%. The Y-axis measures the volume of parameters transmitted by each client during the training process as in the X-axis of Fig. 3. It is observed that `HIST` takes less communication to achieve the preset accuracy, which demonstrates the efficiency of the proposed algorithm over HFedAvg. In addition, as the number of cells increases from 2 to 6, the communication cost shows a decreasing trend, which forms a sharp comparison with HFedAvg. This further demonstrates the advantage of the proposed `HIST` algorithm. However, when we increase the number of cells to 8, `HIST` also suffers performance degradation. This can be attributed to the submodel getting too small for effectively handling the

Convolutional Neural Network on Fashion-MNIST								
	Fully nonn-i.i.d.				Cell i.i.d., client non-i.i.d.			
	2 Cells		4 Cells		2 Cells		4 Cells	
Accuracy	HIST	HFedAvg	HIST	HFedAvg	HIST	HFedAvg	HIST	HFedAvg
70%	13.20	23.66	8.08	20.28	4.40	6.76	3.32	11.84
80%	115.30	206.18	76.06	246.74	38.72	86.20	24.72	89.58

TABLE I: Communication cost (MB) per client for achieving testing accuracy of 70% and 80% on Fashion-MNIST

Convolutional Neural Network on CIFAR-10								
	Fully nonn-i.i.d.				Cell i.i.d., client non-i.i.d.			
	2 Cells		4 Cells		2 Cells		4 Cells	
Accuracy	HIST	HFedAvg	HIST	HFedAvg	HIST	HFedAvg	HIST	HFedAvg
50%	177.50	392.16	110.64	392.16	114.76	227.04	74.72	268.32
70%	856.14	1609.92	528.18	1795.68	584.52	1217.76	394.84	1568.64

TABLE II: Communication cost (MB) per client for achieving testing accuracy of 50% and 70% on CIFAR-10

task[3], highlighting the trade-off between training costs and testing accuracy, which is also consistent with our theoretical results.

Effects of System Parameters: The impacts of period of edge aggregation H and period of global synchronization E on the convergence behavior are demonstrated in Fig. 5. The X-axis represents the number of global synchronizations at the cloud server. We consider a scenario with $N = 3$. As E increases from 5 to 10 to 15, HIST attains a better convergence performance, which is witnessed by both Figs. 5a and 5b. This is because a large E gives rise to more rounds of edge training. When H increases from 20 to 40, and from 40 to 60, Fig. 5b shows that the convergence speed of HIST first enjoys an acceleration and then a degradation, where the latter is induced by data heterogeneity. This phenomenon also fits well with our theory, where we show that there is an upper bound for the number of local updates. Fig. 5b shows that HIST attains a better performance as H increases from 40 to 60. This is because this data distribution exhibits lower data heterogeneity, which allows for a larger number of local updates.

2) Extension to convolutional neural network: For CNNs, following the existing literature [18], [23], we only partition the fully connected layers while the convolutional layers are shared by different cells. Hence, the communication load for each client per global round is $\frac{\text{vol}_c+\text{vol}_f/N}{\text{vol}_c+\text{vol}_f}E$ times of the size of the full model, where vol_c denotes the volume of parameters of convolutional layers and vol_f denotes the volume of parameters of fully connected layers. Since the fully connected layer occupies a significant portion of the entire neural network's parameter volume for many deep models, such a partition strategy still notably benefits the training efficiency [18]. For instance, the total number of parameters in the convolutional layers of LeNet-5 is 2572 while the fully connected layers consist of 59134 parameters. The partitionable parameters are 95.8% of the entire model.

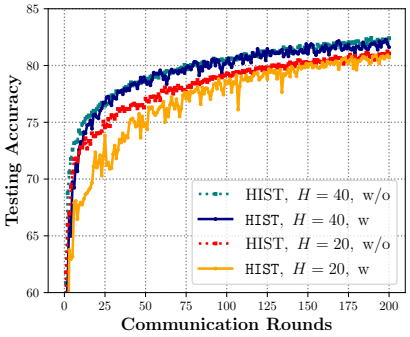
[3]If the number of cells gets too large in specific scenarios, the HIST algorithm can be adapted by modifying the partition strategy (e.g., by dividing the full model with partial overlapping) to prevent performance degradation.

We numerically investigate the performance of HIST by training different CNN models on Fashion MNIST and CIFAR-10. Specifically, for Fashion MNIST, we use LeNet-5; for CIFAR-10, we follow the architecture shown in https://github.com/heysachin/Convolutional-Neural-Network-CIFAR-10 PyTorch/blob/master/cifar10_cnn.ipynb. The size of the former model is 0.169 MB and the latter is 2.064 MB. Under this setup, we compared the communication cost required by the proposed HIST and HFedAvg for attaining some preset accuracies.

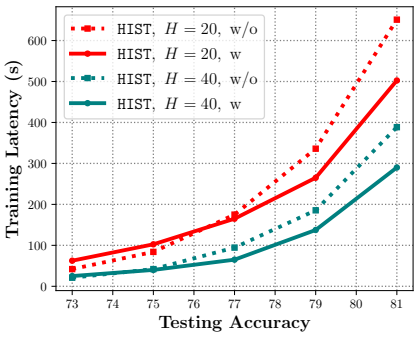
For the Fashion-MNIST, we set the target accuracies to 70% and 80%, while for the CIFAR-10, the target accuracies are set to 50% and 70%. The required communication cost per client for achieving these preset accuracies are shown in Tables I and II. These experiments were conducted with a configuration of $H = 40$ and $E = 5$. It can be observed from both Tables I and II that the proposed HIST takes less communication cost to achieve the preset accuracies than HFedAvg. Moreover, the advantage of the HIST algorithm gets more significant when the number of cells increases from $N = 2$ to $N = 4$. For example, referring to Table II, in the case of training a convolutional neural network model to achieve a 70% accuracy level, the communication cost of the HIST algorithm is $\frac{1}{1.88}$ of what is required by HFedAvg when the number of cells is $N = 2$. Notably, this ratio improves to $\frac{1}{3.4}$ as the number of cells increases to $N = 4$, indicating that the HIST algorithm becomes more advantageous (with a smaller ratio signifying better performance) in scenarios with more cells. This aligns with the phenomena observed in Section VI-B1. Taken together, these findings show that the proposed hierarchical submodel training methodology significantly enhances efficiency in training both the CNN and the fully connected neural network.

C. Effect of Partitioning Optimization in HIST

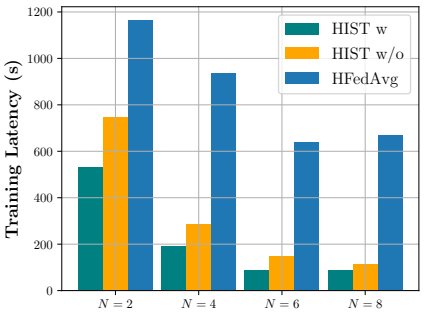
This subsection aims to evaluate the effectiveness of our proposed partition optimization strategy in reducing the training latency. We consider training of a fully connected neural



(a) The impact of mask optimization on the convergence speed.

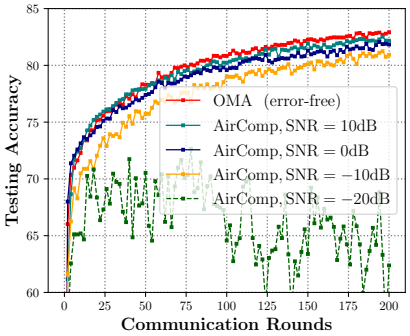


(b) The impact of mask optimization on the training latency.

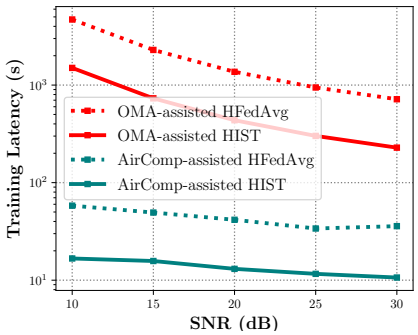


(c) Training latency comparison for attaining accuracy of 80% under different N.

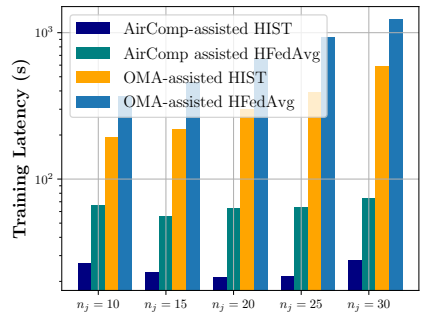
Fig. 6: The impact of mask optimization in HIST. We compare the cases with mask optimization (w) and without mask optimization (w/o).



(a) The impact of SNR on the convergence speed.



(b) The impact of SNR on the training latency.



(c) Training latency comparison for attaining accuracy of 80% under different n_j.

Fig. 7: The performance of AirComp-assisted HIST. The proposed mask optimization strategy combined with AirComp-assisted HIST provide significant performance advantages compared with baselines.

network on Fashion-MNIST under the non-i.i.d. data setup. The model architecture is detailed in Section VI-B1. In the following experiments, we determine the CPU frequency for each client based on their cell location. Specifically, for client $i \in \mathcal{C}_j,\ j \in [1, \frac{N}{2}]$, the CPU frequency is chosen randomly within the range $(1, 2)$ GHz. For clients in cells numbered in the latter half, the frequency range from which we draw is $(2, 4)$ GHz. Additionally, we determine the data rate for each client using the following formula: $\mathcal{R}_i = B\log(1 + \text{SNR}\|\boldsymbol{h}_i^t\|^2)$, in which B represents the bandwidth of the respective cell, SNR is calculated as $\frac{P_i}{\sigma_0^2}$, and $\boldsymbol{h}_i^t$ denotes the channel between client i and the corresponding edge server during the t-th global round. We let bandwidth B be 9 MHz for all cells. Meanwhile, we set SNR to 30 dB for clients $i \in \mathcal{C}_j$, $j \in [1, \frac{N}{2}]$ and 40 dB for for clients $i \in \mathcal{C}_j$, $j \in (\frac{N}{2}, N]$. The channel is modeled as $\boldsymbol{h}_i^t \sim \mathcal{CN}(0, I_M),\ \forall i, t$. In addition, V_0 is set to 10^6 cycles per update and ϵ_{th} is set to $9\delta_1^2$. The number of antennas at each edge server is set to $M = 10$.

In Fig. 6, we compare the performance of the HIST algorithm with and without integration of mask optimization. In this context, 'with mask optimization' is denoted as 'w', while 'uniform partition' is denoted as 'w/o'. This experiment was conducted with $N = 4$ and $E = 5$. As shown in Fig. 6a, when using the optimized mask tailored to training latency minimization, HIST exhibits a slightly slower convergence speed in terms of communication rounds, compared to the case with a uniform partition. Note that uniform partition is considered optimal in terms of the convergence bound but it does not consider the network resource availability of the system. On the other hand, as depicted in Fig. 6b, the optimized mask effectively reduces the training latency of the HIST algorithm across various settings. This is because an optimized partition will not compromise the convergence speed largely based on condition (19c) (as shown in Fig. 6a) but notably reduce per-round training latency.

Fig. 6c compares the training latency of HIST with mask optimization, the HIST with a uniform partitioning, and HFedAvg. The latency of all these algorithms decreases as the number of cells increases from 2 to 6, which can be attributed to the fact that the communication complexity at each edge server linear decreases as the number of clients in the cell decreases. Additionally, when the number of cells increases from 6 to 8, HFedAvg experiences a significant increase in training latency while HIST shows comparatively less degradation. This can be attributed to the fact that the per-round reduction in training latency partially compensates for the degradation in convergence speed. Additionally, it is clear that HIST with partitioning optimization can always beat

these baselines for all $N \in \{2,4,6,8\}$. These observations confirm that the HIST combined with an optimized mask could improve the training efficacy by a wide margin.

D. Performance Evaluation of the AirComp-assisted HIST

This subsection provides numerical results of the AirComp-assisted HIST algorithm. Fig. 7a plots the convergence behavior of the AirComp-assisted HIST Algorithm in training the fully connected neural network on Fashion-MNIST under different SNR values. This experiment was carried out with $N = 4$, $H = 40$, and $E = 5$. We take the OMA-assisted HIST as a benchmark whose convergence behavior (with respect to communication rounds) is independent of the SNR value. The results show that the AirComp-assisted HIST performs well under a wide SNR region, which reveals the robustness of the proposed algorithm. It also can be observed that a low SNR degrades the convergence and leads to a big oscillation. This is because a low SNR leads to a large variance in the model aggregated by AirComp, which may hinder HIST from converging. Such a phenomenon fits well with our theoretical analysis provided in Section V.

Fig. 7b shows the required training latency for attaining the target testing accuracy of 80%. We compare the training latency of the OMA- and AirComp-assisted HFedAvg and HIST under different SNR values. The CPU frequencies of clients are drawn from $(2,4)$ GHz. Other settings are the same as that of Fig. 7a. As shown in Fig. 7b, the training latency monotonically decreases as the SNR increases for all algorithms. For the OMA scheme, an error-free aggregation, the SNR affects the upload rate, which in turn impacts the latency experienced in each round of communication. In the AirComp scheme, a larger SNR results in a smaller error during aggregation and thus reduces the training latency for achieving the preset testing accuracy.

Fig. 7c shows the impact of the number of clients in each cell on the training latency. We consider $N = 4$, $H = 40$, and $E = 5$. The number of clients in each cell is set to $\{10, 15, 20, 25, 30\}$. Other parameters including CPU frequencies and SNRs are the same as the ones in Section VI-C. As depicted in Fig. 7c, the AirComp-assisted HIST significantly reduces latency compared to both the OMA-assisted HIST and the AirComp-assisted HFedAvg. Additionally, it is clear that the latency of the OMA-assisted HIST shows a notably upward trend as the number of clients increases while the fluctuation of the AirComp-assisted algorithms is slighter. This can be attributed to the fact that the communication latency of the OMA schemes linearly increases with the number of clients whereas the per-round latency of the AirComp-assisted algorithms is independent of the number of clients. The latency trend directly reflects the speed of convergence. For the AirComp-assisted HIST, the downward trend from 10 to 20 can be attributed to the speedup of convergence concerning the number of clients at each cell as discussed in Remark 1. On the other hand, more clients will also induce higher data heterogeneity in the data distribution. As the number of clients continues to increase to a specific value, i.e., 25, the effect of adding clients will be dominated by the induced data heterogeneity, thus slowing down the convergence.

VII. Conclusion

In this paper, we proposed a hierarchical federated submodel training algorithm termed HIST, that is efficient in terms of communication, computation, storage, and training time by integrating independent submodel training with local training. We investigated its convergence behavior with arbitrary submodel partitioning under non-convex loss functions and non-i.i.d. data settings, and characterized the impacts of submodel partitioning sizes, non-i.i.d. extent, the number of periods of edge and global aggregations, and the number of cells on the convergence. Based on the derived convergence bound, we proposed to minimize the training latency by designing the model partition strategy. Subsequently, we adopted AirComp to assist local model aggregations within each cell to further enhance the efficiency of HIST over hierarchical wireless networks. As a result, the communication complexity of the global server and edge servers become independent of the number of cells and the number of clients, respectively. Extensive simulation results show that HIST is able to achieve the target accuracy much faster with less training costs compared to the baseline. The improvements in training latency brought by our submodel partitioning optimization and AirComp are also demonstrated empirically.

Appendix

A. Proof of Theorem 1

For analysis, we introduce virtual iterate $\hat{\boldsymbol{x}}^{t,e} = \sum_{j=1}^{N} \bar{\boldsymbol{x}}_j^{t,e}$, which denotes the virtual synchronized global model of the proposed HIST algorithm. For notation ease, we denote

$$\begin{aligned} D_t &= \sum_{j=1}^{N} \sum_{e=0}^{E-1} \mathbb{E} \left\| \hat{\boldsymbol{x}}^{t,e} - \bar{\boldsymbol{x}}_j^{t,e} \right\|^2 \text{ and} \\ Q_t &= \sum_{j=1}^{N} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \frac{1}{H} \sum_{e=0}^{E-1} \sum_{h=0}^{H-1} \mathbb{E} \left\| \bar{\boldsymbol{x}}_j^{t,e} - \boldsymbol{x}_{i,h}^{t,e} \right\|^2 . \end{aligned} \tag{37}$$

The proofs of Theorems 1 and 2 rely on the following three lemmas, which are in turn proven in the supplementary material.

Lemma 1. *Suppose that Assumptions 2-5 hold, $N \geq 2$, and $\gamma \leq \frac{1}{2HL}$, then the iterates generated by HIST algorithm satisfy*

$$\begin{aligned} \mathbb{E}[f(\bar{\boldsymbol{x}}^{t+1})] \leq & \mathbb{E}[f(\bar{\boldsymbol{x}}^{t})] - \frac{\gamma H}{2} \sum_{e=0}^{E-1} \mathbb{E} \left\| \nabla f(\hat{\boldsymbol{x}}^{t,e}) \right\|^2 + \gamma^2 E H \tilde{N} L \sigma^2 \\ & + \frac{3\gamma}{2} \frac{E H N d_{max}}{d} \delta_1^2 + \frac{3 H \gamma L^2}{2} \left(D_t + Q_t \right) . \end{aligned} \tag{38}$$

Lemma 2. *Suppose that Assumptions 2-5 hold, $N \geq 2$, and $\gamma \leq \frac{1}{2HL}$, then the iterates generated by AirComp-assisted HIST algorithm satisfy*

$$\begin{aligned} \mathbb{E}[f(\bar{\boldsymbol{x}}^{t+1})] \leq & \mathbb{E}[f(\bar{\boldsymbol{x}}^{t})] - \frac{\gamma H}{2} \sum_{e=0}^{E-1} \mathbb{E} \left\| \nabla f(\hat{\boldsymbol{x}}^{t,e}) \right\|^2 + \gamma^2 E H \tilde{N} L \sigma^2 \\ & + \frac{3\gamma}{2} \frac{E H N d_{max}}{d} \delta_1^2 + \frac{1}{2} \sum_{e=0}^{E-1} \sum_{j=1}^{N} \mathrm{MSE}_j^{t,e} \\ & + \frac{3 H \gamma L^2}{2} \left(D_t + Q_t \right) . \end{aligned} \tag{39}$$

Lemma 3. *Suppose that Assumptions 2-5 hold and* $\gamma \leq \frac{1}{EL\sqrt{54(N+1)}}$*, then the difference between the edge models and the global model can be bounded as*

$$\begin{aligned} D_t \leq & 162\gamma^2(N-1)E^2H^2\sum_{e=0}^{E-1}\mathbb{E}\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\|^2 + 4E(N-1)\mathbb{E}\left\|\bar{\boldsymbol{x}}^t\right\|^2 \\ & + 18\gamma^2(N+1)E^2H^2L^2Q_t + 6\gamma^2(N+1)E^2H\tilde{N}\sigma^2 \\ & + 108\gamma^2(N+1)E^3H^2\frac{Nd_{max}}{d}\delta_1^2. \end{aligned} \tag{40}$$

Lemma 4. *Suppose that Assumptions 2-4 and 6 hold and* $\gamma \leq \frac{1}{\sqrt{15}HL}$*, then the difference between the local models and the edge models can be bounded as*

$$\begin{aligned} Q_t \leq & 5\gamma^2H^2L^2D_t + 5\gamma^2NH^2\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 \\ & + 2\gamma^2NHE\sigma^2 + 5\gamma^2NH^2E\delta_2^2 + 5\gamma^2NH^2E\delta_1^2. \end{aligned} \tag{41}$$

Lemmas 1 and 2 characterize the dynamics of the global loss function. Lemmas 3 and 4 characterize the upper bound of the diversity between the virtual global model and edge models and between the edge model and local models, respectively.

With Lemmas 1, 3, and 4, we can prove Theorem 1 as follows.

Based on Lemmas 3, 4 and Fact (3), we have

$$\begin{aligned} D_t + Q_t \leq & \tilde{\alpha}\Bigg\{162\gamma^2(N-1)E^2H^2\sum_{e=0}^{E-1}\mathbb{E}\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\|^2 \\ & + 4E(N-1)\mathbb{E}\left\|\bar{\boldsymbol{x}}^t\right\|^2 + 6\gamma^2(N+1)E^2H\tilde{N}\sigma^2 \\ & + 108\gamma^2(N+1)E^3H^2\frac{Nd_{\max}}{d}\delta_1^2\Bigg\} \\ & + \tilde{\beta}\Big\{2\gamma^2NHE\sigma^2 + 5\gamma^2NH^2E\delta_2^2 + 5\gamma^2NH^2E\delta_1^2 \\ & + 5\gamma^2NH^2\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2\Big\}, \end{aligned} \tag{42}$$

where $\tilde{\alpha} = \frac{1+5\gamma^2H^2L^2}{1-18\gamma^2(N+1)E^2H^2L^2*5\gamma^2H^2L^2}$ and $\tilde{\beta} = \frac{1+18\gamma^2(N+1)E^2H^2L^2}{1-18\gamma^2(N+1)E^2H^2L^2*5\gamma^2H^2L^2}$. According to the setting of γ, one can claim $\tilde{\alpha} \leq 2$, $\tilde{\beta} \leq 2$. Additionally, according to Lemma 1, we have

$$\begin{aligned} \sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 \leq & \frac{2}{\gamma H}\mathbb{E}[f(\bar{\boldsymbol{x}}^t)] - \mathbb{E}[f(\bar{\boldsymbol{x}}^{t+1})] + 2\gamma E\tilde{N}L\sigma^2 \\ & + 3\frac{ENd_{\max}}{d}\delta_1^2 + 3L^2(D_t+Q_t). \end{aligned} \tag{43}$$

Plugging (42) into (43), we have

$$\begin{aligned} & \left(1-\gamma^2L^2\left(972(N-1)E^2H^2+30NH^2\right)\right)\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 \\ \leq & 2\frac{\mathbb{E}[f(\bar{\boldsymbol{x}}^t)] - \mathbb{E}[f(\bar{\boldsymbol{x}}^{t+1})]}{\gamma H} + 2\gamma E\tilde{N}L\sigma^2 + 30\gamma^2L^2NH^2E\delta_2^2 \\ & + \gamma^2L^2\left((36(N+1)E^2H\tilde{N}+12NHE\right)\sigma^2 \\ & + \gamma^2L^2\left(648(N+1)\frac{Nd_{\max}}{d}E^3H^2+30NH^2E\right)\delta_1^2 \\ & + 24E(N-1)L^2\mathbb{E}\left\|\bar{\boldsymbol{x}}^t\right\|^2 + 3\frac{ENd_{\max}}{d}\delta_1^2. \end{aligned} \tag{44}$$

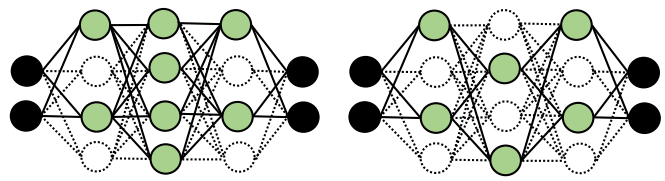
As $1-\gamma^2L^2\left(972(N-1)E^2H^2+30NH^2\right) \leq \frac{1}{2}$, we obtain

$$\begin{aligned} & \frac{1}{E}\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 \leq 4\frac{\mathbb{E}[f(\bar{\boldsymbol{x}}^t)] - \mathbb{E}[f(\bar{\boldsymbol{x}}^{t+1})]}{\gamma EH} \\ & + \left(4\gamma\tilde{N}L + \gamma^2L^2\left((72(N+1)EH\tilde{N}+24NH\right)\right)\sigma^2 \\ & + \gamma^2L^2\left(1296(N+1)\frac{Nd_{\max}}{d}E^2H^2+60NH^2\right)\delta_1^2 \\ & + 60\gamma^2L^2NH^2\delta_2^2 + 6\frac{Nd_{\max}}{d}\delta_1^2 + 48(N-1)L^2\mathbb{E}\left\|\bar{\boldsymbol{x}}^t\right\|^2. \end{aligned} \tag{45}$$

Based on the setting of γ in (13), we can obtain Theorem 1.

With Lemmas 2, 3, and 4, we can prove Theorem 2 in the same way as detailed above for Theorem 1.

B. Model partition details



(a) Our developed model partition scheme, partition ratio $\frac{1}{N}$

(b) Proposed by [18], , partition ratio $\frac{1}{N^2}$

Fig. 8: Comparison of partition strategies for a neural network with multiple hidden layers

Theoretically, we need to first generate masks satisfying (3) and then partition the model as shown in (3). In experiments, we, following the existing works [18], [23], turn to partition the neurons, such that all the resulting submodels are fully connected neural networks, which is better supported by the existing ML computing libraries like PyTorch and TensorFlow. The difference between our partition scheme and the one proposed by [18] is demonstrated in Fig. 8, where the partition ratio is defined as the proportion of the size of each submodel to the size of the entire model. Specifically, we let the input and output neurons be independent of partition and partition the hidden neurons every two layers. As a result, the parameter volume of each submodel is equal to $1/N$ of that of the full model on average. [18] proposed to partition the hidden neurons at every layer, which is more aggressive but could deteriorate the performance and result in non-robustness. In particular, when the number of hidden layers is 1, such as the model used in Section VI-B1, our strategy is the same as that proposed in [18]. The advantage of our scheme can be reflected in training LeNet-5, considered in Section VI-B2. As mentioned in [18], they rely on some extra tricks to train this model, or IST does not work. However, with our partition scheme, IST works well under the regular training routine.

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C. Supplementary Material: Proof of Lemmas

Before proving the above lemmas, we introduce some notations. Denote $\mathbb{E}_t^e$ as an expectation conditioned on the historical information up to the start of the (t,e)-th round. Let $\boldsymbol{g}_{i,h}^{t,e}$ denote the stochastic gradient $\nabla l(\boldsymbol{x}_{i,h}^{t,e}, \xi_{i,h}^{t,e})$, $\xi_{i,h}^{t,e} \sim \mathcal{D}_i$. In addition, we present three facts that will be used in this subsection.

Fact 1. *Suppose that mask $\boldsymbol{p}_j$ is randomly generated via rule (3) and $\|\boldsymbol{p}_j\|_1 = d_j$, then*

$$\mathbb{E}[\boldsymbol{p}_j \odot \boldsymbol{z}] = \frac{d_j}{d}\|\boldsymbol{z}\|^2,\ \forall j \text{ and } \sum_{j=1}^{N} \left\|\boldsymbol{p}_j^t \odot (\nabla f(\boldsymbol{x}) - \nabla f_j(\boldsymbol{x}))\right\|^2 \leq \frac{N d_{max}}{d}\delta_1^2, \forall \boldsymbol{x},$$

where $d_{max} = \max\{d_1, d_2, \ldots, d_N\}$.

Proof. The former can be derived by the following series of transformations

$$\mathbb{E}[\|\boldsymbol{p}_j \odot \boldsymbol{z}\|^2] = \mathbb{E}[\sum_{k=1}^{d}((p_j)_k z_k)^2] = \mathbb{E}[\sum_{k=1}^{d}((p_j)_k z_k)^2] = \sum_{k=1}^{d}\mathbb{E}[((p_j)_k z_k)^2] = \sum_{k=1}^{d}\frac{d_j}{d}z_k^2 = \frac{d_j}{d}\|\boldsymbol{z}\|^2,$$

where $(p_j)_k$ and z_k represent the k-th elements of $\boldsymbol{p}_j$ and $\boldsymbol{z}$, respectively. Furthermore, combining it with Assumption 5 gives rise to the latter. □

Fact 2. *Any masks $\{\boldsymbol{p}_j\}_{j=1}^N$ generated by rule (3) satisfy*

$$\sum_{j=1}^{N}\|\boldsymbol{p}_j \odot \boldsymbol{z} - \boldsymbol{z}\|^2 = (N-1)\|\boldsymbol{z}\|^2.$$

Proof.

$$\sum_{j=1}^{N}\|\boldsymbol{p}_j \odot \boldsymbol{z} - \boldsymbol{z}\|^2 = \sum_{j=1}^{N}\{\|\boldsymbol{p}_j \odot \boldsymbol{z}\| - 2\langle \boldsymbol{p}_j \odot \boldsymbol{z}, \boldsymbol{z}\rangle + \|\boldsymbol{z}\|^2\} = (N-1)\|\boldsymbol{z}\|^2.$$

□

Fact 3. *For the following inequalities,*

$$\begin{aligned} x &\leq \alpha y + a \\ y &\leq \beta x + b, \end{aligned} \tag{46}$$

where $\alpha\beta < 1$, we have $x + y \leq \frac{1+\beta}{1-\alpha\beta}a + \frac{1+\alpha}{1-\alpha\beta}b$.

1) Proof of Lemma 1: Based on the virtual iteration $\hat{\boldsymbol{x}}^{t,e+1} = \hat{\boldsymbol{x}}^{t,e} - \gamma\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,e}$ and Assumption 2, we have

$$\mathbb{E}_t^e[f(\hat{\boldsymbol{x}}^{t,e+1})] \leq f(\hat{\boldsymbol{x}}^{t,e}) \underbrace{-\gamma\mathbb{E}_t^e\left\langle \nabla f(\hat{\boldsymbol{x}}^{t,e}), \sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,e}\right\rangle}_{T_1} + \gamma^2 L \underbrace{\frac{1}{2}\mathbb{E}_t^e\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,e}\right\|^2}_{T_2}. \tag{47}$$

Resorting to Cauchy-Schwartz inequality, we bound T_2 as

$$T_2 \leq \underbrace{\mathbb{E}_t^e\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,e} - \sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right\|^2}_{T_2^l} + \mathbb{E}_t^e\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right\|^2. \tag{48}$$

Denoting $\boldsymbol{a}_i = \sum_{h=0}^{H-1}\left(\boldsymbol{g}_{i,h}^{t,e} - \nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right)$, we have

$$\begin{aligned} T_2^l &= \sum_{j=1}^{N}\mathbb{E}_t^e\left\|\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\boldsymbol{a}_i\right\|^2 = \sum_{j=1}^{N}\frac{1}{n_j^2}\sum_{i\in\mathcal{C}_j}\mathbb{E}_t^e\left\|\boldsymbol{p}_j^t \odot \boldsymbol{a}_i\right\|^2 \leq \sum_{j=1}^{N}\frac{1}{n_j^2}\sum_{i\in\mathcal{C}_j}\mathbb{E}_t^e\left\|\boldsymbol{a}_i\right\|^2 \\ &= \sum_{j=1}^{N}\frac{1}{n_j^2}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\mathbb{E}_t^e\left\|\boldsymbol{g}_{i,h}^{t,e} - \nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right\|^2 \leq H\tilde{N}\sigma^2, \end{aligned} \tag{49}$$

where the first equality holds because there is no overlapping between any two different masks in the same round, the second equality follows the fact that $\mathbb{E}_t^e[\boldsymbol{a}_i]=0$ and $\boldsymbol{a}_i$ is independent of $\boldsymbol{a}_j$ for any $j\neq i$, the third equality follows [47, Lemma 2], and the final inequality comes from Assumption 4. We thus obtain

$$T_2 \leq H\tilde{N}\sigma^2 + \mathbb{E}_t^e\left\|\sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right\|^2. \tag{50}$$

Utilizing $\mathbb{E}_t^e\left[\sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,e} - \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right]=0$, we rewrite T_1 as follows

$$\begin{aligned}T_1 =& -\gamma H\mathbb{E}_t^e\left\langle \nabla f(\hat{\boldsymbol{x}}^{t,e}), \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\frac{1}{H}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right\rangle \\ =& \frac{\gamma H}{2}\left\{\underbrace{\mathbb{E}_t^e\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e}) - \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\frac{1}{H}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right\|^2}_{T_3} - \left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 - \mathbb{E}_t^e\left\|\sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\frac{1}{H}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right\|^2\right\}.\end{aligned} \tag{51}$$

Based on the facts that $\nabla f(\hat{\boldsymbol{x}}^{t,e}) = \sum_{j=1}^N \boldsymbol{p}_j^t \odot \nabla f(\hat{\boldsymbol{x}}^{t,e})$ and $\|\sum_{j=1}^N \boldsymbol{p}_j^t \odot \boldsymbol{z}_j\|^2 = \sum_{j=1}^N \|\boldsymbol{p}_j^t \odot \boldsymbol{z}_j\|^2$, we can bound T_3 as

$$\begin{aligned}T_3 =& \sum_{j=1}^N \mathbb{E}_t^e\left\|\boldsymbol{p}_j^t \odot \left(\nabla f(\hat{\boldsymbol{x}}^{t,e}) - \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\frac{1}{H}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right)\right\|^2 \\ \leq& \frac{1}{H}\sum_{h=0}^{H-1}\underbrace{\sum_{j=1}^N \mathbb{E}_t^e\left\|\boldsymbol{p}_j^t \odot \left(\nabla f(\hat{\boldsymbol{x}}^{t,e}) - \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right)\right\|^2}_{T_3'},\end{aligned} \tag{52}$$

where the inequality follows Jensen's inequality. Inserting $\mp\nabla f_j(\hat{\boldsymbol{x}}^{t,e}) \mp \nabla f_j(\bar{\boldsymbol{x}}_j^{t,e})$ into T_3' and calling Cauchy-Schwartz inequality, we have

$$\begin{aligned}T_3' \leq& 3\sum_{j=1}^N \mathbb{E}_t^e\left\|\boldsymbol{p}_j^t \odot \left(\nabla f(\hat{\boldsymbol{x}}^{t,e}) - \nabla f_j(\hat{\boldsymbol{x}}^{t,e})\right)\right\|^2 + 3\sum_{j=1}^N \mathbb{E}_t^e\left\|\boldsymbol{p}_j^t \odot \left(\nabla f_j(\hat{\boldsymbol{x}}^{t,e}) - \nabla f_j(\bar{\boldsymbol{x}}_j^{t,e})\right)\right\|^2 \\ &+ 3\sum_{j=1}^N \mathbb{E}_t^e\left\|\boldsymbol{p}_j^t \odot \left(\nabla f_j(\bar{\boldsymbol{x}}_j^{t,e}) - \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right)\right\|^2.\end{aligned} \tag{53}$$

Given Fact 1, it follows that

$$3\sum_{j=1}^N \left\|\boldsymbol{p}_j^t \odot \left(\nabla f(\hat{\boldsymbol{x}}^{t,e}) - \nabla f_j(\hat{\boldsymbol{x}}^{t,e})\right)\right\|^2 \leq 3\frac{Nd_{\max}}{d}\delta_1^2. \tag{54}$$

In addition, recalling Jensen's inequality, we have

$$\begin{aligned}\sum_{j=1}^N \mathbb{E}_t^e\left\|\boldsymbol{p}_j^t \odot \left(\nabla f_j(\bar{\boldsymbol{x}}_j^{t,e}) - \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right)\right\|^2 =& \sum_{j=1}^N \mathbb{E}_t^e\left\|\boldsymbol{p}_j^t \odot \left(\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\bar{\boldsymbol{x}}_j^{t,e}) - \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right)\right\|^2 \\ \leq& \sum_{j=1}^N \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}_t^e\left\|\nabla F_i(\bar{\boldsymbol{x}}_j^{t,e}) - \nabla F_i(\boldsymbol{x}_{i,h}^{t,e})\right\|^2.\end{aligned} \tag{55}$$

Combining (52), (53), (54), and (55), and then utilizing the smoothness of F_i and f_j defined in Assumption 2, we have

$$T_3 = 3\frac{Nd_{\max}}{d}\delta_1^2 + 3L^2\sum_{j=1}^N \mathbb{E}_t^e\left\|\hat{\boldsymbol{x}}^{t,e} - \bar{\boldsymbol{x}}_j^{t,e}\right\|^2 + 3L^2\sum_{j=1}^N \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\frac{1}{H}\sum_{h=0}^{H-1}\mathbb{E}_t^e\left\|\bar{\boldsymbol{x}}_j^{t,e} - \boldsymbol{x}_{i,h}^{t,e}\right\|^2. \tag{56}$$

Combining (59), (50), (51), and (56), we obtain

$$\begin{aligned}\mathbb{E}_t^e[f(\hat{\boldsymbol{x}}^{t,e+1})] \leq & f(\hat{\boldsymbol{x}}^{t,e}) - \frac{\gamma H}{2}\mathbb{E}_t^e\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 + \gamma^2 H\tilde{N}L\sigma^2 + \frac{3\gamma H}{2}\frac{Nd_{\max}}{d}\delta_1^2 \\ & + \frac{3\gamma HL^2}{2}\left\{\sum_{j=1}^N \mathbb{E}_t^e\left\|\hat{\boldsymbol{x}}^{t,e} - \bar{\boldsymbol{x}}_j^{t,e}\right\|^2 + \sum_{j=1}^N \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\frac{1}{H}\sum_{h=0}^{H-1}\mathbb{E}_t^e\left\|\bar{\boldsymbol{x}}_j^{t,e} - \boldsymbol{x}_{i,h}^{t,e}\right\|^2\right\}.\end{aligned} \tag{57}$$

Taking an expectation over all the randomness for the above inequality and telescoping it from $e = 0$ to $e = H - 1$, we will obtain Lemma 1.

2) Proof of Lemma 2: With AirComp, the iteration becomes

$$\hat{\boldsymbol{x}}^{t,e+1} = \hat{\boldsymbol{x}}^{t,e} - \gamma\sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,e} + \sum_{j=1}^N \boldsymbol{p}_j^t \odot \boldsymbol{n}_j^{t,e}, \tag{58}$$

Note that $\boldsymbol{p}_j^t \odot \boldsymbol{n}_j^{t,e} = \boldsymbol{n}_j^{t,e}$. According to Assumption 2, we have

$$\begin{aligned}\mathbb{E}_t^e[f(\hat{\boldsymbol{x}}^{t,e+1})] \leq & f(\hat{\boldsymbol{x}}^{t,e}) \underbrace{-\gamma\mathbb{E}_t^e\left\langle\nabla f(\hat{\boldsymbol{x}}^{t,e}), \sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,e} - \sum_{j=1}^N \boldsymbol{p}_j^t \odot \boldsymbol{n}_j^{t,e}\right\rangle}_{G_1} \\ & + L\underbrace{\frac{1}{2}\mathbb{E}_t^e\left\|\gamma\sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,e} - \sum_{j=1}^N \boldsymbol{p}_j^t \odot \boldsymbol{n}_j^{t,e}\right\|^2}_{G_2}.\end{aligned} \tag{59}$$

Since $\boldsymbol{n}_j^{t,e}$ is independent of $\boldsymbol{g}_{i,h}^{t,e}$ and $\mathbb{E}[\boldsymbol{n}_j^{t,e}] = 0$, the upper bound of G_1 is the same as T_1 in (59). Additionally, we can rewrite G_2 as

$$G_2 = \frac{1}{2}\gamma^2\mathbb{E}_t^e\left\|\sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,e}\right\|^2 + \frac{1}{2}\sum_{j=1}^N \mathbb{E}\|\boldsymbol{p}_j^t \odot \boldsymbol{n}_j^{t,e}\|^2. \tag{60}$$

Next, following the same step in the proof of Lemma 3, we can obtain

$$\begin{aligned}\mathbb{E}_t^e[f(\hat{\boldsymbol{x}}^{t,e+1})] \leq & f(\hat{\boldsymbol{x}}^{t,e}) - \frac{\gamma H}{2}\mathbb{E}_t^e\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 + \gamma^2 H\tilde{N}L\sigma^2 + \frac{3\gamma H}{2}\frac{Nd_{\max}}{d}\delta_1^2 + \frac{1}{2}\sum_{j=1}^N \mathrm{MSE}_j^{t,e} \\ & + \frac{3\gamma HL^2}{2}\left\{\sum_{j=1}^N \mathbb{E}_t^e\left\|\hat{\boldsymbol{x}}^{t,e} - \bar{\boldsymbol{x}}_j^{t,e}\right\|^2 + \sum_{j=1}^N \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\frac{1}{H}\sum_{h=0}^{H-1}\mathbb{E}_t^e\left\|\bar{\boldsymbol{x}}_j^{t,e} - \boldsymbol{x}_{i,h}^{t,e}\right\|^2\right\},\end{aligned} \tag{61}$$

where $\mathrm{MSE}_j^{t,e} = \mathbb{E}\|\boldsymbol{p}_j^t \odot \boldsymbol{n}_j^{t,e}\|^2$. Taking an expectation over all the randomness for the above inequality and telescoping it from $e = 0$ to $e = H - 1$, we will obtain Lemma 2.

3) Proof of Lemma 3: According to the iteration in Algorithm 1, we have

$$\begin{aligned}\mathbb{E}\left\|\hat{\boldsymbol{x}}^{t,e} - \bar{\boldsymbol{x}}_j^{t,e}\right\|^2 = & \mathbb{E}\left\|\bar{\boldsymbol{x}}^t - \gamma\sum_{\tau_1=0}^{e-1}\sum_{j=1}^N \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,\tau_1} - \bar{\boldsymbol{x}}_j^{t,0} + \gamma\sum_{\tau_1=0}^{e-1}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,\tau_1}\right\|^2 \\ \leq & 2\mathbb{E}\left\|\bar{\boldsymbol{x}}^t - \bar{\boldsymbol{x}}_j^{t,0}\right\|^2 + 2\gamma^2\underbrace{\mathbb{E}\left\|\sum_{j=1}^N \boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,\tau_1} - \boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\boldsymbol{g}_{i,h}^{t,\tau_1}\right\|^2}_{T_{4,j}^{t,e}}\end{aligned} \tag{62}$$

For $T_{4,j}^{t,e}$, we have

$$
\begin{aligned}
T_{4,j}^{t,e} =& \mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\left(\boldsymbol{g}_{i,h}^{t,\tau_1} \mp \nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1})\right) - \boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\left(\boldsymbol{g}_{i,h}^{t,\tau_1} \mp \nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1})\right)\right\|^2 \\
\leq& \underbrace{3\mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\left(\boldsymbol{g}_{i,h}^{t,\tau_1} - \nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1})\right)\right\|^2 + 3\mathbb{E}\left\|\boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\left(\boldsymbol{g}_{i,h}^{t,\tau_1} - \nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1})\right)\right\|^2}_{T_{5,j}^{t,e}} \\
&+ \underbrace{3\mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1}) - \boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1})\right\|^2}_{T_{6,j}^{t,e}}.
\end{aligned}
\tag{63}
$$

For $T_{5,j}^{t,e}$, we have

$$
\begin{aligned}
T_{5,j}^{t,e} =& 3\sum_{j=1}^{N}\mathbb{E}\left\|\boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\left(\boldsymbol{g}_{i,h}^{t,\tau_1} - \nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1})\right)\right\|^2 + 3\mathbb{E}\left\|\boldsymbol{p}_j^t \odot \sum_{\tau_1=0}^{e-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\left(\boldsymbol{g}_{i,h}^{t,\tau_1} - \nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1})\right)\right\|^2 \\
\leq& 3\sum_{j=1}^{N}\sum_{\tau_1=0}^{e-1}\frac{1}{n_j^2}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\mathbb{E}\left\|\boldsymbol{g}_{i,h}^{t,\tau_1} - \nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1})\right\|^2 + 3\sum_{\tau_1=0}^{e-1}\frac{1}{n_j^2}\sum_{i\in\mathcal{C}_j}\sum_{h=0}^{H-1}\mathbb{E}\left\|\boldsymbol{g}_{i,h}^{t,\tau_1} - \nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1})\right\|^2 \\
\leq& 3eH\tilde{N}\sigma^2 + 3eH\frac{1}{n_j}\sigma^2,
\end{aligned}
\tag{64}
$$

where the first inequality comes from [47, Lemma 2] and the inequality $\|\boldsymbol{p}\odot\boldsymbol{z}\| \leq \|\boldsymbol{z}\|^2$ and the second inequality follows Assumption 4.

For $T_{6,j}^{t,e}$, we have

$$
\begin{aligned}
T_{6,j}^{t,e} =& 9eH\sum_{\tau_1=0}^{e-1}\sum_{h=0}^{H-1}\mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\left(\nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1}) \mp \nabla F_i(\bar{\boldsymbol{x}}_j^{t,\tau_1})\right) - \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\left(\nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1}) \mp \nabla F_i(\bar{\boldsymbol{x}}_j^{t,\tau_1})\right)\right\|^2 \\
\leq& 9eH\sum_{\tau_1=0}^{e-1}\sum_{h=0}^{H-1}\mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\left(\nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1}) - \nabla F_i(\bar{\boldsymbol{x}}_j^{t,\tau_1})\right)\right\|^2 + 9eH\sum_{\tau_1=0}^{e-1}\sum_{h=0}^{H-1}\mathbb{E}\left\|\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\left(\nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1}) - \nabla F_i(\bar{\boldsymbol{x}}_j^{t,\tau_1})\right)\right\|^2 \\
&+ 9eH^2\sum_{\tau_1=0}^{e-1}\mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\bar{\boldsymbol{x}}_j^{t,\tau_1}) - \boldsymbol{p}_j^t \odot \frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\nabla F_i(\bar{\boldsymbol{x}}_j^{t,\tau_1})\right\|^2 \\
\leq& 9eH\sum_{\tau_1=0}^{e-1}\sum_{h=0}^{H-1}\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1}) - \nabla F_i(\bar{\boldsymbol{x}}_j^{t,\tau_1})\right\|^2 + 9eH\sum_{\tau_1=0}^{e-1}\sum_{h=0}^{H-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\nabla F_i(\boldsymbol{x}_{i,h}^{t,\tau_1}) - \nabla F_i(\bar{\boldsymbol{x}}_j^{t,\tau_1})\right\|^2 \\
&+ 9eH^2\sum_{\tau_1=0}^{e-1}\mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \nabla f_j(\bar{\boldsymbol{x}}_j^{t,\tau_1}) - \boldsymbol{p}_j^t \odot f_j(\bar{\boldsymbol{x}}_j^{t,\tau_1})\right\|^2 \\
\leq& 9eHL^2\sum_{\tau_1=0}^{e-1}\sum_{h=0}^{H-1}\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{x}_{i,h}^{t,\tau_1} - \bar{\boldsymbol{x}}_j^{t,\tau_1}\right\|^2 + 9eHL^2\sum_{\tau_1=0}^{e-1}\sum_{h=0}^{H-1}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{x}_{i,h}^{t,\tau_1} - \bar{\boldsymbol{x}}_j^{t,\tau_1}\right\|^2 \\
&+ 9eH^2\sum_{\tau_1=0}^{e-1}\underbrace{\mathbb{E}\left\|\sum_{j=1}^{N}\boldsymbol{p}_j^t \odot \nabla f_j(\bar{\boldsymbol{x}}_j^{t,\tau_1}) - \boldsymbol{p}_j^t \odot f_j(\bar{\boldsymbol{x}}_j^{t,\tau_1})\right\|^2}_{T_{7,j}^{t,e}}.
\end{aligned}
\tag{65}
$$

$$
\begin{aligned}
T_{7,j}^{t,e} =& \mathbb{E}\left\|\sum_{j=1}^{N} \boldsymbol{p}_j^t \odot \left(\nabla f_j(\bar{\boldsymbol{x}}_j^{t,\tau_1}) \mp \nabla f_j(\hat{\boldsymbol{x}}^{t,\tau_1})\right) - \boldsymbol{p}_j^t \odot \left(\nabla f_j(\bar{\boldsymbol{x}}_j^{t,\tau_1}) \mp f_j(\hat{\boldsymbol{x}}^{t,\tau_1})\right)\right\|^2 \\
\leq& 3\mathbb{E}\left\|\sum_{j=1}^{N} \boldsymbol{p}_j^t \odot \left(\nabla f_j(\bar{\boldsymbol{x}}_j^{t,\tau_1}) - \nabla f_j(\hat{\boldsymbol{x}}^{t,\tau_1})\right)\right\|^2 + 3\left\|\boldsymbol{p}_j^t \odot \left(\nabla f_j(\bar{\boldsymbol{x}}_j^{t,\tau_1}) - \nabla f_j(\hat{\boldsymbol{x}}^{t,\tau_1})\right)\right\|^2 \\
&+ 3\mathbb{E}\left\|\sum_{j=1}^{N} \boldsymbol{p}_j^t \odot \nabla f_j(\hat{\boldsymbol{x}}^{t,\tau_1}) - \boldsymbol{p}_j^t \odot \nabla f_j(\hat{\boldsymbol{x}}^{t,\tau_1})\right\|^2 \\
\leq& 3L^2 \sum_{j=1}^{N} \mathbb{E}\left\|\bar{\boldsymbol{x}}_j^{t,\tau_1} - \hat{\boldsymbol{x}}^{t,\tau_1}\right\|^2 + 3L^2\left\|\bar{\boldsymbol{x}}_j^{t,\tau_1} - \hat{\boldsymbol{x}}^{t,\tau_1}\right\|^2 \\
&+ \underbrace{3\mathbb{E}\left\|\sum_{j=1}^{N} \boldsymbol{p}_j^t \odot \left(\nabla f_j(\hat{\boldsymbol{x}}^{t,\tau_1}) \mp \nabla f(\hat{\boldsymbol{x}}^{t,\tau_1})\right) - \boldsymbol{p}_j^t \odot \left(\nabla f_j(\hat{\boldsymbol{x}}^{t,\tau_1}) \mp \nabla f(\hat{\boldsymbol{x}}^{t,\tau_1})\right)\right\|^2}_{T_{8,j}^{t,e}}
\end{aligned} \tag{66}
$$

Similarly, we further bound $\sum_{j=1}^{N} T_{8,j}^{t,e}$ as follows

$$
\begin{aligned}
\sum_{j=1}^{N} T_{8,j}^{t,e} \leq& 9(N+1)\sum_{j=1}^{N} \mathbb{E}\left\|\boldsymbol{p}_j^t \odot \left(\nabla f_j(\hat{\boldsymbol{x}}^{t,\tau_1}) - \nabla f(\hat{\boldsymbol{x}}^{t,\tau_1})\right)\right\|^2 + 9\sum_{j=1}^{N} \mathbb{E}\left\|\boldsymbol{p}_j^t \odot \nabla f(\hat{\boldsymbol{x}}^{t,\tau_1}) - \nabla f(\hat{\boldsymbol{x}}^{t,\tau_1})\right\|^2 \\
\leq& 9(N+1)\frac{N d_{\max}}{d}\delta_1^2 + 9(N-1)\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,\tau_1})\right\|^2,
\end{aligned} \tag{67}
$$

where the second inequality comes from Facts 1, 2 and Assumption 5.

$$
\begin{aligned}
\sum_{e=0}^{E-1}\sum_{j=1}^{N} T_{4,j}^{t,e} \leq& \frac{3}{2}(N+1)\tilde{N}E^2 H\sigma^2 + \frac{9}{2}(N+1)E^2 HL^2 \sum_{e=0}^{E-1}\sum_{h=0}^{H-1}\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{x}_{i,h}^{t,e} - \bar{\boldsymbol{x}}_j^{t,e}\right\|^2 \\
&\frac{27}{2}(N+1)E^2H^2L^2\sum_{e=0}^{E-1}\sum_{j=1}^{N}\mathbb{E}\left\|\bar{\boldsymbol{x}}_j^{t,e} - \hat{\boldsymbol{x}}^{t,e}\right\|^2 \\
&27(N+1)E^3H^2\frac{N d_{\max}}{d}\delta_1^2 + \frac{81}{2}(N-1)E^2H^2\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2.
\end{aligned} \tag{68}
$$

Plugging (68) into (62) and utilizing Fact 2, we have

$$
\begin{aligned}
D_t \leq& 2E(N-1)\mathbb{E}\left\|\bar{\boldsymbol{x}}^t\right\|^2 + 3\gamma^2(N+1)E^2H\tilde{N}\sigma^2 + 9\gamma^2(N+1)E^2H^2L^2Q_t \\
&+ 27\gamma^2(N+1)E^2H^2L^2D_t + 54\gamma^2(N+1)E^3H^2\frac{N d_{\max}}{d}\delta_1^2 \\
&+ 81\gamma^2(N-1)E^2H^2\sum_{e=0}^{E-1}\mathbb{E}\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\|^2.
\end{aligned} \tag{69}
$$

As $1 - 27\gamma^2(N+1)E^2H^2L^2 \geq \frac{1}{2}$ when $\gamma \leq \frac{1}{\sqrt{54(N+1)}EHL}$, we thus obtain Lemma 3.

4) Proof of Lemma 4: Based on the iteration $\bar{\boldsymbol{x}}_j^{t,e} - \boldsymbol{x}_{i,h}^{t,e} = \gamma \boldsymbol{p}_j^t \sum_{\tau_2=0}^{h-1} \boldsymbol{g}_{i,\tau_2}^{t,e}$, we have

$$
\begin{aligned}
\mathbb{E}\left\|\bar{\boldsymbol{x}}_j^{t,e} - \boldsymbol{x}_{i,h}^{t,e}\right\|^2 \leq& 2\gamma^2\mathbb{E}\left\|\sum_{\tau_2=0}^{h-1}\boldsymbol{g}_{i,\tau_2}^{t,e} - \sum_{\tau_2=0}^{h-1}\nabla F_i(\boldsymbol{x}_{i,\tau_2}^{t,e})\right\|^2 + 2\gamma^2\mathbb{E}\left\|\sum_{\tau_2=0}^{h-1}\nabla F_i(\boldsymbol{x}_{i,\tau_2}^{t,e})\right\|^2 \\
=& 2\gamma^2\sum_{\tau_2=0}^{h-1}\mathbb{E}\left\|\boldsymbol{g}_{i,\tau_2}^{t,e} - \nabla F_i(\boldsymbol{x}_{i,\tau_2}^{t,e})\right\|^2 + 2\gamma^2\mathbb{E}\left\|\sum_{\tau_2=0}^{h-1}\nabla F_i(\boldsymbol{x}_{i,\tau_2}^{t,e})\right\|^2 \\
\leq& 2\gamma^2 h\sigma^2 + 2\gamma^2 h\sum_{\tau_2=0}^{h-1}\mathbb{E}\left\|\nabla F_i(\boldsymbol{x}_{i,\tau_2}^{t,e})\right\|^2,
\end{aligned} \tag{70}
$$

where the first inequality follows Cauchy-Schwartz inequality, the equality comes from [47, Lemma 2], and the second inequality follows Assumption 4. Additionally, we can bound $\mathbb{E}\left\|\nabla F_i(\boldsymbol{x}_{i,\tau_2}^{t,e})\right\|^2$ as

$$\begin{aligned}\mathbb{E}\left\|\nabla F_i(\boldsymbol{x}_{i,\tau_2}^{t,e})\right\|^2 =&\mathbb{E}\left\|\nabla F_i(\boldsymbol{x}_{i,\tau_2}^{t,e}) \mp \nabla F_i(\bar{\boldsymbol{x}}_j^{t,e}) \mp \nabla f_j(\bar{\boldsymbol{x}}_j^{t,e}) \mp \nabla f(\bar{\boldsymbol{x}}_j^{t,e}) \mp \nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2\\ \leq&5\mathbb{E}\left\|\nabla F_i(\boldsymbol{x}_{i,\tau_2}^{t,e}) - \nabla F_i(\bar{\boldsymbol{x}}_j^{t,e})\right\|^2 + 5\mathbb{E}\left\|\nabla F_i(\bar{\boldsymbol{x}}_j^{t,e}) - \nabla f_j(\bar{\boldsymbol{x}}_j^{t,e})\right\| + 5\mathbb{E}\left\|\nabla f_j(\bar{\boldsymbol{x}}_j^{t,e}) - \nabla f(\bar{\boldsymbol{x}}_j^{t,e})\right\|^2\\ &+ 5\mathbb{E}\left\|\nabla f(\bar{\boldsymbol{x}}_j^{t,e}) - \nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 + 5\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2\\ \leq&5L^2\mathbb{E}\left\|\boldsymbol{x}_{i,\tau_2}^{t,e} - \bar{\boldsymbol{x}}_j^{t,e}\right\|^2 + 5\mathbb{E}\left\|\nabla F_i(\bar{\boldsymbol{x}}_j^{t,e}) - \nabla f_j(\bar{\boldsymbol{x}}_j^{t,e})\right\| + 5\mathbb{E}\left\|\nabla f_j(\bar{\boldsymbol{x}}_j^{t,e}) - \nabla f(\bar{\boldsymbol{x}}_j^{t,e})\right\|^2\\ &+ 5L^2\mathbb{E}\left\|\bar{\boldsymbol{x}}_j^{t,e} - \hat{\boldsymbol{x}}^{t,e}\right\|^2 + 5\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2,\end{aligned} \tag{71}$$

where the first inequality comes from Cauchy-Schwartz inequality and the second one follows Assumption 2. Hence, we have

$$\begin{aligned}\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\bar{\boldsymbol{x}}_j^{t,e} - \boldsymbol{x}_{i,\tau_2}^{t,e}\right\|^2 \leq&10\gamma^2 hL^2\sum_{\tau_2=0}^{h-1}\sum_{j=1}^{N}\frac{1}{n_j}\sum_{i\in\mathcal{C}_j}\mathbb{E}\left\|\boldsymbol{x}_{i,\tau_2}^{t,e} - \bar{\boldsymbol{x}}_j^{t,e}\right\|^2 + 10\gamma^2h^2L^2\sum_{j=1}^{N}\mathbb{E}\left\|\bar{\boldsymbol{x}}_j^{t,e} - \hat{\boldsymbol{x}}^{t,e}\right\|^2\\ &+ 10\gamma^2Nh^2\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 + 2\gamma^2Nh\sigma^2 + 10\gamma^2Nh^2\delta_2^2 + 10\gamma^2Nh^2\delta_1^2.\end{aligned} \tag{72}$$

Recalling the definitions of Q_t and Q_t in (37), we have

$$Q_t \leq 5\gamma^2H^2L^2Q_t + \frac{10}{3}\gamma^2H^2L^2D_t + \frac{10}{3}\gamma^2NH^2\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla f(\hat{\boldsymbol{x}}^{t,e})\right\|^2 + \gamma^2NHE\sigma^2 + \frac{10}{3}\gamma^2NH^2E\delta_2^2 + \frac{10}{3}\gamma^2NH^2E\delta_1^2. \tag{73}$$

As $3\left(1-5\gamma^2H^2L^2\right) \geq 2$ when $\gamma \leq \frac{1}{\sqrt{15}HL}$, we thus obtain Lemma 4.